# Patent Exposure and the Diffusion of "Trapped Knowledge"

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Valuable knowledge developed in one part of the world may remain trapped even when formally available because frictions prevent it from entering the consideration set of downstream users. This paper examines whether granting US patents to already-established foreign inventions can "untrap" this knowledge by increasing its effective exposure, making it more likely to be noticed, recognized as relevant, and seriously evaluated. Using difference-in-differences with an examiner leniency instrument, I find that US grants of foreign patents significantly increase both the intensity and reach of forward citations. Using a new measure of "trappedness," I show that knowledge from historically more trapped countries and sectors experiences larger diffusion benefits after US grants. These findings highlight the historical role of the US as an influential platform of global knowledge diffusion.

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# I. Introduction

Not all valuable knowledge finds its way to where it is needed. While the common metaphor of "standing on the shoulders of giants" captures the cumulative nature of innovation, it overlooks a critical challenge: the visible giants may not be the only ones worth standing on. Knowledge originating from certain countries, institutions, or languages may circulate widely and become foundational, while other potentially valuable ideas remain confined to local or regional contexts. As a result, innovations developed in one part of the world may remain "trapped" locally: not because they lack merit, but because frictions limit their global exposure and integration. Such frictions can arise from linguistic barriers, differences in institutional norms, and structural biases that favor knowledge from already dominant innovation hubs.

This paper explores how increasing the exposure of local knowledge affects global innovation. I focus on the United States, whose patent system is globally visible and influential and whose institutional features allow for a clean causal analysis. In particular, I examine whether receiving a US patent can help disseminate foreign innovations that are already established in their home countries. Although foreign inventors primarily seek US patents to protect commercial interests in a large market, a US grant may also increase an invention's effective exposure. The innovations in my sample have already been granted in their home countries and are publicly available before the US application, so the US grant does not provide first disclosure. Instead, becoming an issued US patent may increase effective exposure by making the invention more salient, easier to recognize as relevant, and more credible to downstream innovators, increasing the likelihood that the already-available knowledge is noticed, evaluated, and ultimately built upon.

To study this effect, I focus on patent families with granted patents in non-US home countries that subsequently file applications through the USPTO. This sample is especially informative because it allows me to observe the dynamic effects following home-country grants but prior to the issuance of US grants. I, therefore, compare the diffusion trajectories of foreign patents that receive a US grant with those that do not. A key concern is that the US grant status may be endogenous to observed or unobserved characteristics like patent quality, which may also drive the diffusion of knowledge and there-

fore bias the estimates. To address this, I instrument the US grant status with examiner leniency, exploiting quasi-random assignment to stricter versus more lenient examiners. This shifts grant outcomes for reasons orthogonal to the application's characteristics such as intrinsic quality. In an ideal world where publicly available knowledge receives equal exposure regardless of its origin or institutional setting, this estimated effect would be small, as the underlying invention is already disclosed and should receive similar exposure with or without a US grant; the effect could even be negative if US protection raises legal costs that deter follow-on use.

The analysis proceeds in several stages. First, I estimate the impact of receiving a US patent grant on forward citations using the panel of foreign-origin patent families. The results show that receiving a US patent grant significantly increases forward citations: US grants increase the likelihood of receiving any forward citation by about 17% and raise the total number of citations by roughly 27%. The dynamic effects plot shows no meaningful difference in citation trajectories in the years leading up to the grants. Because the underlying inventions are already publicly disclosed before grant, these findings are consistent with US grants increasing their effective exposure rather than merely making the technical information available.

If the US functions as a platform for global exposure, then one would expect the diffusion benefits to appear in regions beyond the United States and the invention's country of origin. Therefore, I examine how the impact spreads across different geographies. In addition to sizeable increases in citations within the US and the home country, citations from third countries rise markedly by about 23%, indicating that US grants may help extend knowledge to places that are otherwise more disconnected from its origin. Further, US grants extend the knowledge to more middle- and low-income countries, hinting that these countries may face greater informational barriers and therefore benefit more from the amplified exposure provided by the US platform. In this sense, US grants not only enhance awareness in the US innovation system, but also act as a bridge that carries knowledge across borders to new, and often more peripheral, regions.

Next, I show that the increased exposure associated with a US grant extends beyond the focal invention to the inventor's broader portfolio of patents. I find that citations to

the inventors' other patents increase significantly after the US grants, with the strongest effects for less established inventors with a limited number of patents. This portfolio-level boost suggests that US grants may help elevate an entire line of work, not just a single invention, by drawing international attention.

Further, I explore why some inventions gain more citations than others after US grants. A key factor is "trappedness": the extent to which knowledge from a particular country or sector receives limited exposure within the global innovation network. I use exposure broadly to capture whether knowledge that is formally available is sufficiently visible, salient, and credible to enter the consideration set of downstream innovators. Limited exposure can therefore arise either because an invention is not sufficiently noticed or because its relevance or technical value is difficult to assess. A US grant may help untrap such knowledge by increasing its exposure along these margins. I find that the effects are concentrated among patents from countries with historically limited global reach, such as Korea and China, while patents from English-dominant or highly integrated countries show little change. Importantly, language barriers alone do not fully account for these patterns, as US grants also lead to a sizable increase in citations from foreign countries that share the home-country language. To better quantify the trappedness of each country and sector, I then develop a new citation-based measure based on whether patents from certain countries and technology sectors are cited by inventors or only surfaced later by examiners. The intuition is that examiner-added citations reveal relevant knowledge that was not readily recognized or brought into inventors' consideration sets. Using this measure, I find that US grants have a larger effect when the originating country is more trapped. This relationship also holds across sectors, consistent with US grants playing a particularly important role in increasing the exposure of knowledge that might otherwise remain under-recognized. Taken together, these findings suggest that US patent grants serve as an influential platform amplifying the exposure of valuable but less utilized knowledge, particularly when those ideas originate from places that are less integrated into the global innovation system.

Finally, I examine whether the citation response could reflect channels other than increased exposure to trapped knowledge. First, citations may reflect legal and proce-

dural citation incentives: applicants may cite granted US patents to satisfy disclosure obligations or reduce litigation risks, while examiners may cite them to assess novelty, reject overlapping claims, or clarify patent boundaries. The effect, however, remains strong for citations to the home-priority patent, which are less directly tied to US legal rights, and for citations excluding examiner-added US citations. Second, the grant may facilitate commercialization, so downstream innovators may encounter the invention through products, licensing, exports, or commercial relationships rather than through patent awareness itself. Yet, the effect persists among patents classified as non-product-oriented, where direct product-market interaction is less likely. Moreover, product-market interaction typically takes time to develop, whereas the citation response appears almost immediately after the US grant.

Taken together, these findings speak to several strands of literature on cumulative innovation and knowledge diffusion. The idea that innovation builds cumulatively on prior knowledge is foundational. In endogenous economic growth models, past ideas function as non-rivalrous inputs into future invention, enabling sustained economic growth (Romer, 1990; Jones, 1995). A rich empirical literature explores the cumulative nature of innovation and how it is shaped by geographic and institutional frictions (Jaffe et al., 1993; Galasso and Schankerman, 2015; Furman and Stern, 2011). Beyond the production of knowledge, the accessibility of knowledge is also important. Mokyr (2011) emphasizes that the economic significance of useful knowledge hinges not only on its production but on its accessibility. The ability of a society to stand on the shoulders of giants depends on the quality of mechanisms for storing, certifying, and accessing that knowledge (Furman and Stern, 2011; Wagner et al., 2014). Yet formal accessibility does not guarantee equal exposure: the country or institutional origin of an idea can shape the attention it receives even conditional on its intellectual content or quality (Morgan et al., 2018; Gomez et al., 2022; Fry and MacGarvie, 2024; Qiu et al., 2025; Nagaraj and Yao, 2026). This paper extends this literature by studying whether institutions can help integrate potentially valuable knowledge that remains confined to local contexts into the broader cumulative innovation process.

A parallel literature in international economics and innovation studies highlights the

barriers to cross-border knowledge diffusion. Classic work emphasizes how technological, geographic, historical, and institutional factors play critical roles in the spread of ideas and technologies across national boundaries (Eaton and Kortum, 1996; Keller, 2004; Spolaore and Wacziarg, 2012; Rogers et al., 2014). Further empirical studies document such frictions in cross-border citation, licensing behavior, and their productivity consequences (Branstetter, 2001; MacGarvie, 2005, 2006; Montobbio et al., 2015; De Rassenfosse and Seliger, 2020; Berkes et al., 2022; Miguelez et al., 2026). Much of the literature on overcoming these barriers focuses on individuals as carriers of knowledge across borders (Kerr, 2008; Borjas and Doran, 2012; Ganguli, 2015; Kahn and MacGarvie, 2016; Fry, 2023; Fry and Furman, 2025; Fry and Ganguli, 2026). This paper shifts attention from mobile individuals to mobile ideas, emphasizing how institutional recognition by the USPTO may increase the exposure of valuable foreign-origin knowledge and facilitate its diffusion to otherwise disconnected regions.

The findings also speak to long-standing debates about the role of intellectual property rights in shaping innovation and diffusion. On one side, intellectual property rights are seen as beneficial for encouraging investment, creating markets for ideas, and bringing more knowledge into the public domain (Gans and Stern, 2000; Hellman, 2007; Merges and Nelson, 1994). Recent causal studies show that patents can facilitate diffusion by accelerating disclosure, supporting markets for ideas, improving translation and disclosure quality, and releasing inventions from secrecy (Hegde and Luo, 2018; Baruffaldi and Simeth, 2018; Büttner et al., 2022; Hegde et al., 2023; de Rassenfosse et al., 2020; Gross, 2023; Dyer et al., 2024; Higham and Nagaoka, 2026), while patent grants may also certify quality to external audiences (Conti et al., 2013). Recent papers also examine how first entrance into the US by receiving US patent grants can increase knowledge flows and growth (Farre-Mensa et al., 2020; Gong et al., 2023; de Grazia et al., 2026). On the other side, the anti-commons perspective warns that stronger IP protections may block follow-on research by making knowledge harder to access (Heller and Eisenberg, 1998; David, 2004; Gaessler et al., 2025). Empirical studies also show that patenting can have zero or negative effects on follow-on citations (Murray and Stern, 2007; Sampat and Williams, 2019; Williams, 2013). The evidence in this paper points to another channel through which patents may facilitate knowledge flows: by increasing the effective exposure of

already-disclosed but potentially overlooked foreign ideas.

Relatedly, the results intersect with a growing literature on the appropriateness of technology in global contexts. This literature highlights that even high-potential innovations may fail to spread if they are not perceived as suitable or relevant across different settings (Moscona and Sastry, 2025; Lerner et al., 2024). This paper raises a complementary concern: even when innovations are appropriate, they may not reach broader audiences when their exposure is limited. The increase in attention from countries well beyond the origin and the US, particularly from other middle- and low-income countries, suggests that US grants may help expose appropriate technologies to more potential users.

The remainder of this article proceeds as follows. Section II provides background on trapped knowledge and the role of patents in knowledge diffusion. Section III describes the data and empirical strategy. Section IV presents the main results and explores the underlying mechanisms. Section V concludes.

# II. Background

## A. Trapped Knowledge in Global Innovation

Across sectors and disciplines, many innovations, practices, or discoveries remain underused despite their potential to generate societal benefits. For instance, Israel's water-saving drip irrigation systems offered considerable promise for agriculture in water-scarce regions, but adoption lagged in many countries due to the lack of exposure, training, or adaptation mechanisms. Similarly, India's high-volume, low-cost cataract surgery model had the potential to reduce blindness in many low-income countries, but knowledge of the approach remained localized for years due to limited diffusion across health systems.

There are many reasons why useful technologies remain underused: lack of awareness, institutional barriers, geopolitical fragmentation, high costs, and more. In some cases, limited adoption may be rational when local conditions make adoption less beneficial or more costly. However, in other cases, valuable knowledge fails to spread not because it lacks relevance or utility, but because it lacks exposure. I refer to this as "trapped knowledge": knowledge that remains geographically confined despite its global potential,

primarily because of limited exposure. One illustrative case of geographically trapped knowledge comes from China's agricultural sector.

### *Case Study: China's Hybrid Rice*

In the 1960s, Yuan Longping discovered the genetic basis of heterosis in rice, where the offspring of two genetically distinct parents exhibit superior traits. This finding was notable because heterosis had been thought impossible in self-pollinating crops like rice. Building on this, Chinese researchers bred high-yield, resilient hybrid varieties in the 1970s that became central to China's food strategy, boosting output and reducing hunger. By 1994, hybrid rice accounted for 57% of national rice production with average yields in some provinces reaching 36–50% above the conventional 5 metric tons per hectare benchmark (Yuan, 1998).

Despite its success in China, this innovation remained largely unknown and less utilized outside the country, especially in Africa, where it is potentially well-suited. African countries like Nigeria, Senegal, and Madagascar have very high rice consumption, but local yields around Sub-Saharan Africa averaged just 2.4 tons per hectare due to poor seed quality, erratic rainfall, and limited inputs, making up only 6% of global total rice production with 26% of total harvested area (FAOSTAT, 2022; Yuan et al., 2024). China's hybrids matched these constraints: they are high-yielding; tolerant to heat, drought, and poor soils; and some lines exhibit resistance to threats such as bacterial blight. Unlike US or European varieties optimized for large-scale, mechanized production under controlled irrigation, Chinese hybrids often require hand-planting of young seedlings and were developed for rain-fed, labor-intensive, smallholder systems common across African regions. Despite this suitability, few efforts were made to introduce it to Africa due to language barriers, institutional disconnects, and a development landscape dominated by Western actors. African governments often looked to institutions like the International Rice Research Institute (IRRI) or the World Bank, but largely overlooked Chinese advances.

Changes began in the 1990s, when Chinese researchers affiliated with US institutions initiated influential hybrid rice research (Yu et al., 1997), which in turn spurred follow-on studies in the United States. In the early 2000s, Chinese institutions began working

with African partners on hybrid rice demonstration and training. The Africa Rice Center launched a hybrid rice program in partnership with the Chinese Academy of Agricultural Sciences and IRRI under the Green Super Rice initiative (El-Namaky and Demont, 2013). Countries like Côte d'Ivoire, Liberia, Madagascar, Mozambique, Nigeria, Tanzania, and Uganda began evaluating Chinese hybrids (Zenna et al., 2017). These efforts yielded real gains: for instance, Chinese hybrid rice has allowed Tanzania to become the largest regional producer, showing a fivefold increase in paddy rice exports in 2015 compared with the early 2000s (Siméon et al., 2022).

Appendix Figure A1 Panel (a) shows the rise of hybrid rice research in China, the US, and other middle- and low-income countries based on publication counts. Defining the "emergence year" as the point when a region reaches 10% of its total hybrid rice publications, hybrid rice research emerged in China in 1983, in the US in 1999, and in other mid- or low-income countries around 2005. While many factors may explain this sequence, one plausible mechanism is that the growing body of hybrid rice research in the US during the 90s may have facilitated greater attention, adoption, and local research in other developing regions. Panel (b) is consistent with this idea: hybrid-rice patents authored by Chinese researchers tend to reach broader audiences when they are filed abroad, primarily in the US. Those filed in the US receive nearly twice as many forward citations on average as those filed in China, and have a much higher share of citations coming from a diverse set of regions.

Taken together, Chinese hybrid rice provides an illustrative example of "trapped knowledge" being untrapped. Once localized agricultural expertise was finally diffused to the right context, it addressed a long-standing productivity gap, improved food security, and reduced import dependence for countries struggling to meet domestic rice demand. Besides the value of the technology itself, this example highlights the importance of creating mechanisms for knowledge to diffuse to places where it is needed. In this regard, some evidence suggests that the US may play a role in helping locally developed knowledge to diffuse globally. Building on this intuition, this paper focuses on the US as a channel through which geographically confined knowledge may be untrapped and brought into broader circulation.

## B. Patents and Knowledge Diffusion

Patents, a major form of intellectual property, serve as an important tool in knowledge protection and dissemination. They are designed to protect inventions by granting inventors exclusive rights in exchange for publicly disclosing the underlying knowledge. In principle, the global spread of digital patent databases and translation technologies should make all patents widely searchable. In this sense, patents provide not only a legal shield but also a formal channel through which technical knowledge can, at least in principle, circulate across borders.

Yet in practice, many innovations remain underused, especially those originating from less prominent or non-Western countries. People may overlook these patents due to language barriers, unfamiliar legal systems, or skepticism about their quality. As a result, valuable inventions may remain confined within local markets, preventing their broader diffusion to other regions that could benefit from them (Jaffe et al., 1993; Thompson and Fox-Kean, 2005). This highlights an important distinction between formal disclosure and effective exposure. Public disclosure makes technical knowledge available, but availability does not ensure that potential users notice it, recognize its relevance, or regard it as sufficiently credible to build on it further.

In reality, the motivation for inventors to patent abroad is primarily commercial: inventors seek to protect their competitive position in foreign markets where their products may be sold, produced, or utilized. Filing patents in key jurisdictions like the United States is often a strategic move to facilitate market entry and commercial exploitation. However, international patenting can also affect the global diffusion of the underlying knowledge by increasing its exposure. In the United States, the prominence of the patent system may make a foreign invention more salient to downstream innovators, while successful examination may provide institutional validation that makes the invention easier to evaluate. In this sense, a US grant may increase the effective exposure of knowledge that was already formally available. Admittedly, research also shows that patent rights may impede follow-on innovation by restricting reuse, as emphasized by the anti-commons hypothesis (Murray and Stern, 2007; Sampat and Williams, 2019; Williams, 2013). To the extent that stronger patent rights deter follow-on use, this force works

against observing a positive diffusion response following the grant.

In this paper, patent citations are used as a proxy for knowledge diffusion. Patent citations are imperfect,[1] but well-established as a measure of knowledge diffusion. Survey evidence indicates that citations are systematically associated with communication and learning between inventors, while comparisons with direct measures of firms' knowledge sources show that citations capture codified knowledge flows particularly well (Jaffe et al., 2000; Roach and Cohen, 2013; Jaffe and De Rassenfosse, 2019). At the same time, citations do not capture all forms of learning, particularly knowledge transmitted through private, informal, or contractual channels. The citation outcomes should therefore be interpreted as observable traces of downstream technological reliance rather than as an exhaustive measure.

# III. Data and Methodology

This paper studies the effect of receiving a US patent grant among foreign-origin inventions that seek US protection. It does not claim that the US patent system is uniquely important relative to other major patent offices, which may also facilitate knowledge diffusion through similar exposure-enhancing mechanisms. Instead, it uses the unique institutional setting of the United States to study whether the additional exposure associated with a US patent grant causally increases the global diffusion of foreign-origin knowledge. The data and empirical strategy used for this purpose are described below.

## A. Data

The unit of analysis is patent families: sets of patent applications and publications for the same technology across different countries. The earliest application in a family is the priority application, and the patent office where it is filed is the priority authority (or home country).

I use patent data from PATSTAT Global, which is published by the European Patent Office and contains information on patent applications and granted patents collected from

[1] For instance, Lampe (2012) documents strategic withholding of relevant prior art, although Kuhn et al. (2023) find that this conclusion is sensitive to how withholding is measured.

national and regional patent offices worldwide (EPO, 2023). For each application, the data record the application authority (country), whether the application was granted, corresponding publication information, and the citation network. I then complement these data with bulk data from the USPTO, which provides information on examiners and art units for all US applications. Art units are groups of patent examiners who specialize in particular areas of technology.

Appendix Figure A2 shows the data construction process. The main sample consists of patent families with a granted priority patent in a non-US country and a subsequent US application.[2] I select the families whose subsequent US application was made between 2001 and 2019. This window accommodates two institutional features central to the empirical strategy. First, USPTO patent applications filed on or after November 29, 2000 are published in the public record regardless of whether they are eventually granted as part of the American Inventors Protection Act of 1999. Second, 2019 is the final complete filing year before the USPTO introduced automated routing in October 2020, a reform that substantially increased examiner specialization and effectively eliminated the quasi-random assignment of applications to examiners (Aneja et al., 2024; Pairolero and de Grazia, 2026).

A key sample restriction for testing my hypothesis is that the home-country application must be *granted*, and must be granted *before* the corresponding US application is filed. This restriction focuses the analysis on foreign knowledge that has already been formally recognized and publicly disclosed before entering the US patent system. It also rules out the home-country granting process as a confounder when estimating the effect of US grants, and creates a window in which I can track citation dynamics after the home-country grant but before the US grant.

I further restrict the analysis to patent families with filings only in the home country and the US. Including patent families with filings in additional offices could confound the estimated effect of US grants. If those additional filings are granted on or after the US grant, the estimated US-grant effect may also capture the positive effect of protection in

[2]Patent families with applications filed in no longer existing countries are excluded from the sample. This includes Czechoslovakia, the German Democratic Republic, the Soviet Union, and Yugoslavia/Serbia and Montenegro.

other regions. If they are granted before the US grant, the US-grant effect may instead be underestimated because outcomes may have already increased before the US grant. Note that the results are robust to removing this restriction.

The final dataset consists of 2,470 patent families with outcomes observed over 23 calendar years from 2000 to 2022, corresponding to 506 art units and 19 non-US priority authorities.[3] Admittedly, this sample is small relative to the universe of patents, but it is not hand-curated and instead includes all patent families that satisfy the restrictions needed to test the hypothesis.

Figure 1 shows the distribution of priority authorities and the average grant rate of the US applications from these authorities. East Asian authorities like South Korea, China, and Japan lead the distribution, followed by Western countries such as Germany, Australia, France, and the United Kingdom.

## B. Empirical Strategy

The goal is to estimate how receiving a US patent grant affects follow-on innovation. The unit of analysis is a patent family-year, so a simple comparison can be done to compare two patent families that were originally filed and granted in the same non-US country, belong to the same sector, and filed a US application in the same year, but where only one of the two patent families' US applications is granted. In an ideal world where formally available knowledge receives equal exposure regardless of where it originates or which patent system recognizes it, this difference should be negligible: the underlying invention is already publicly disclosed, so receiving a US grant should not materially change its exposure. The effect could instead be negative if US grants create additional legal barriers that hinder follow-on innovation.

[3] In addition to the standard 4-digit numeric codes (e.g., 2131, 2873, 1611) that belong to Technology Centers, this also includes administrative offices (e.g., Office of Enrollment and Discipline) and special handling divisions (e.g., LO115, EDAD, CSMD).

## 1. Difference-in-Differences

I use a difference-in-differences (DID) framework to estimate the effect of US patent grants. For patent family $i$ at calendar year $t$ with US filing in year $y$, the specification is

$$\text{Citation}_{it} = \alpha + \beta \text{Post Grant}_{it} + \lambda_i + \gamma_{ty(i)} + \varepsilon_{it}, \tag{1}$$

where Post Grant$_{it}$ is an indicator equal to 1 in years after the US grant of treated patent families. $\lambda_i$ are patent family fixed effects, and $\gamma_{ty(i)}$ are the filing cohort $\times$ calendar year fixed effects. While patent family fixed effects remove time-invariant characteristics, how the citations accumulate is highly correlated with the "age" of the patent, which may also correlate with the US grant status. So I include filing cohort $\times$ calendar year fixed effects to allow for cohort-specific time trends. Identification, therefore, comes from within-cohort $\times$ year contrasts: among patent families with their US applications filed in the same year and observed in the same calendar year, those already granted are compared to those not yet granted. The standard errors are clustered by technology class.

Further, I extend the baseline DID specification to a dynamic framework to examine the effects of US patent grants relative to the year of the US grant decision. Specifically, I estimate the following regression:

$$\begin{aligned}\text{Citation}_{it} =& \alpha + \sum_{k=-4, k\neq -1}^{6} \beta_k \cdot \mathbf{1}\{k \text{ years relative to US Grant}\}_{ty(i)} \\ &+ \lambda_i + \gamma_{ty(i)} + \varepsilon_{it}.\end{aligned} \tag{2}$$

The coefficients on the lead years (years before US grants), $\beta_{-4}$ through $\beta_{-2}$, allow me to assess how outcomes respond to future changes in patent grant status. The coefficients on the lag years (years after US grants), $\beta_0$ through $\beta_6$, capture how the estimated effect evolves over time, shedding light on whether the effect is short-lived or persistent. The year immediately before the US grant ($k = -1$) is omitted.

## 2. Examiner Instrument

OLS may yield biased estimates since the probability of being granted is endogenous to the patent's observed and unobserved characteristics.[4] To address this, I follow recent studies that use examiner leniency as an instrumental variable (IV) for patent grant, exploiting quasi-random assignment of examiners as a source of exogenous variation (Sampat and Williams, 2019; Aneja et al., 2024).

Examiners are assigned to patent applications in a manner that is plausibly quasi-random within art units (i.e., based on the last digit of the sequentially assigned application number), and examiners retain discretion in evaluating patentability despite the uniform mandate to grant patents to eligible, novel, non-obvious, and useful inventions (Cockburn et al., 2003; Lemley and Sampat, 2012). This examiner discretion generates substantial variations in outcomes: applications assigned to more lenient examiners are more likely to be granted than similar applications assigned to stricter examiners. Because of the quasi-random assignment, examiner leniency is plausibly uncorrelated with unobserved characteristics of the application, conditional on covariates such as application year and technology class. The IV approach thus identifies a local average treatment effect (LATE) for marginal applications: those whose outcome depends on the leniency of the assigned examiner.

I implement a two-sample two-stage least squares (TS2SLS) estimator following Sampat and Williams (2019) and Angrist and Krueger (1992). For each patent examiner, I retrieve the historical record of all patents handled by that examiner. I then construct the examiner's leniency based on this separate sample as the leave-one-out patent grant rate of an examiner for each application filing year, following Aneja et al. (2024). Specifically, for each US application for patent family $i$ filed in year $y$ and examined by patent examiner $k$ in art unit $a$, the leniency is calculated as

$$\text{Leniency}_{iy(i)k(i)a(i)} = \left(\frac{1}{N_{y(i)k(i)a(i)}}\right)\left(\sum_{l\neq i}^{N_{y(i)k(i)a(i)}} Grant_l\right), \tag{3}$$

[4]Note that all patent families in the treatment group or the control group file a US application, so the selection bias into application (i.e., an applicant's decision to apply for a patent in the US is endogenous to both the applicant's and the patent's observed and unobserved characteristics), is not part of the estimand in the empirics and therefore not considered in this context.

where $N_{y(i)k(i)a(i)}$ is the total number of applications examined by examiner $k$ in filing year $y$ in art unit $a$, and $Grant_l$ equals one if the (non-focal) application is granted by the examiner.[5]

The instrument appears to satisfy the relevance, exclusion, and monotonicity assumptions required for identifying a local average treatment effect. Cross-sectionally, a 10 percentage point increase in examiner leniency is associated with a 9.16 percentage point increase in the likelihood of being granted (SE = 0.052), supporting instrument relevance. The exclusion restriction is plausible given the quasi-random assignment of examiners within art units, making leniency unlikely to be correlated with unobserved application characteristics.[6] Further, monotonicity is also reasonable in this context, because assignment to a more lenient examiner is unlikely to reduce an application's probability of being granted; examiner discretion operates in one direction, primarily by approving marginal cases rather than reversing clearly patentable ones. Appendix Figure A3 shows evidence consistent with the relevance and exclusion assumptions. Examiner leniency is strongly associated with actual grant rates, supporting instrument relevance, but the weak relationship with grant rates predicted from characteristics fixed at application (including technology class, year, priority authority, patent-family size, and number of inventors) provides evidence consistent with exogeneity.

The endogenous variable $Post\ US\ Grant_{it}$ can be viewed as the interaction of a post-period indicator $Post_{it}$ and an indicator for granted families $Treated_i$, and as a result, is instrumented by $Post_{it} \times Leniency_i$. The assumption is that the endogenous part of the treatment is whether a patent will ever be granted or not, instead of the timing of treatment. Namely, conditional on being granted a patent, the exact time when it will be granted is fairly random. To empirically assess this assumption, I use the predicted grant rates based on *ex-ante* characteristics, as discussed above, to capture patent value (Sampat and Williams, 2019; Kline et al., 2019). Appendix Figure A4 shows little association between the predicted grant rate and the timing of the grant, indicating no

[5]The main results are robust to alternative definitions of examiner leniency, i.e., defining the leniency at the examiner-art unit or the examiner-year level. In practice, over 85% of examiners in the sample are only associated with one art unit in a given year.

[6]Righi and Simcoe (2019) document examiner specialization even within art units. I show that the results are robust when restricting the sample to sectors with limited evidence of such specialization.

evidence that applications that appear more likely to be patented based on covariates that are fixed at the time of filing are differentially more likely to be granted earlier. To further address concerns about endogenous timing, I also construct a predicted post-grant indicator $\widehat{Post}_{it} = \mathbf{1}\{t > \widehat{Grant\,Year}_i\}$. Specifically, I predict a grant date for *all* the patents based on their filing date, the assigned art unit, and the leniency of the examiner. All these characteristics are pre-determined and are therefore orthogonal to observed and unobserved characteristics like patent quality. The alternative instrument is then $\widehat{Post}_{it} \times Leniency_i$. The robustness of the results under this alternative method provides additional support for the empirical strategy.

# IV. Empirical Results

The sample of interest includes annual observations from 2000 to 2022 for 2,470 patent families, of which 1,582 ultimately receive a US grant. Appendix Table A1 shows the distribution of CPC classes (3-digit) in the sample.

Summary statistics in Table 1 show that citation outcomes rise notably following the US grant. The average number of forward citations nearly doubles, from 0.72 to 1.40, and the probability of being cited increases from 31% to 53%. Citations from the US exhibit the sharpest shift, with both citation likelihood and counts roughly doubling post-grant. Third-country citations also increase meaningfully, from 0.16 to 0.28 citations per year on average, while home-country citations show a more modest rise. Metrics of citation breadth also show that the number of distinct third countries citing the patent doubles post-grant (from 0.07 to 0.14), with gains observed across both high-income and middle/low-income countries. Finally, measures of average "nonfocal" citations, which are citations to other patents held by the same inventors, also rise post-grant (from 1.05 to 1.70), especially those coming from US and third-country patents.

Taken together, these descriptive patterns suggest that receiving a US patent grant is associated with an increased number of forward citations, possibly through greater global exposure for foreign-origin inventions. The increases extend beyond direct citations to the focal patent family to the geographic breadth of citations and attention to the inventors' broader portfolio. While these differences do not on their own establish causality, they

motivate the following analyses of how US grants affect knowledge diffusion.

The empirical analysis proceeds in several stages. First, I estimate the effect of receiving a US patent grant on forward citations and examine the dynamic effects to assess pre-trends and temporal shifts following the grant. Next, I explore where the additional citations originate, changes in the number and composition of citing countries, and spillovers to other patents by the same inventors. Finally, I investigate whether the effects of US grants are larger for countries and sectors with more "trapped" knowledge, using both group-level comparisons and a citation-based measure of trappedness.

## A. Main Results

Figure 2 plots the average normalized number of citations received in each lead and lag year relative to the US *filing* year, separately for patent families whose US applications are ultimately granted versus those that are never granted.[7] The citations are normalized to mitigate the truncation problem: more recent cohorts of patents will be mechanically less cited, even though they are not less innovative, and the truncation problem varies across technology classes (Lerner and Seru, 2022). I therefore divide the raw number of citations by the average number of citations received by patents from the same filing cohort and patent class.

On and before the filing year, there is no noticeable difference in citation levels between the two groups, suggesting that pre-existing trends do not drive the citation gap. Also, there is little difference in the first year after filing, corresponding well with the fact that the majority of the patents take two years or longer to be granted. Starting from the second year after US filing, however, a clear divergence emerges: granted families begin to accumulate more citations. The difference is relatively stable over time, indicating that, on average, families with granted US applications receive 0.2–0.3 more normalized citations per year. This correspondence between grant timing and citation divergence provides suggestive evidence that it is the US grant itself, rather than other unobserved factors, that drives the observed citation advantage.

[7] Applications are never categorically rejected, but abandoned by applicants following non-final rejections issued by patent examiners (Lemley and Sampat, 2008). Therefore, there is no clear "rejection date" for failed US applications.

Table 2 presents the baseline IV estimates. The results show that receiving a US patent grant leads to a statistically significant increase in forward citations: the probability of receiving any citation increases by 7.2 percentage points (a 23% increase over the pre-grant mean), and the average number of citations rises by 0.27 (a 38% increase). The normalized outcomes, which account for variation across filing year and technology class, show comparable estimates: a 17% increase in citation likelihood and a 27% increase in citation counts.

Figure 3 presents the dynamic effects with three lead years and six lag years relative to the year immediately before US grants. The coefficients on the lead terms show little evidence of difference in citations prior to the US grant, which is inconsistent with the concern that higher-quality patents are both more likely to be cited early and more likely to be granted. Following the grant, citations increase almost immediately, with the effect peaking around two to three years after the grant before gradually tapering off. This pattern suggests that the grant triggers a short-run boost rather than a permanent shift in citation dynamics.

Overall, these results suggest that US patent grants may amplify the exposure of foreign-origin inventions. The effects are consistent across raw and normalized outcomes and robust to a range of alternative specifications. The identification strategy helps address concerns about endogeneity, and the event-study plots confirm that the citation increase coincides closely with the timing of the grant. Taken together, the findings are consistent with granted inventions becoming more integrated into the global innovation system.

### The Diffusion of Knowledge

If US grants increase the global exposure of foreign inventions, an important question is not only whether citations rise, but where those citations come from and how far the knowledge travels. To understand how US patent grants shape the geographic diffusion of foreign-origin inventions, this subsection disaggregates the citation effects by the location of citing inventors, the diversity of countries reached, and the broader spillovers to the inventors' other patents.

Table 3 disaggregates the citation effects of US patent grants by the geographic location of the citing patents.[8] Citations from the US show the strongest response: receiving US grants increases the probability of being cited by US patents by 27% and raises the number of citations by 31%. Citations from the home country also increase, though the effect is more modest, around 20%. This smaller response is consistent with the idea that home-country inventors are already familiar with the invention through local databases, networks, or parallel filings. Interestingly, citations from third countries, those outside both the US and the home country, also rise significantly by 20–23%. These gains are consistent with US grants increasing the exposure of foreign-origin patents among broader international audiences that may otherwise devote limited attention to, or have difficulty evaluating, the underlying knowledge.

Motivated by this sizable increase in citations from third countries, Table 4 examines whether US grants expand not only the volume of follow-on citations but also their geographic breadth. US grants increase the overall number of citing countries by 0.12, a 35% rise relative to the pre-grant mean. Looking at the third countries, US grants lead to a 37% gain in the number of citing countries, driven by the significant 39% increase in the number of citing mid- and low-income countries. Although the magnitude of the effect remains relatively modest in absolute terms, the increase is both statistically significant and economically meaningful. Importantly, the pattern hints that many home-country inventions may be relevant and appropriate for use in third-country settings, but face structural diffusion barriers that a US grant helps overcome. Appendix Figure A6 shows that the citing and cited country-classes tend to be closer in technological composition—measured by the Euclidean distance between their CPC subclass distributions—when the corresponding US application is granted. This difference is highly statistically significant (two-sample t-test of difference in means: $P < 0.0001$) and provides suggestive evidence that US patent grants facilitate the diffusion of innovations to regions where the underlying knowledge is particularly well-suited.

Table 5 approaches the question of diffusion from a different angle: rather than

[8]Differences in the number of observations are due to normalization. If all outcomes are zero for patents in a given US filing year and technology class, the normalization denominator is zero, and the normalized outcome is therefore missing.

focusing on citations to the focal patent family, it examines whether the grant increases downstream attention to other patents from the same inventors. This allows me to assess whether US grants raise the broader scientific profile of the inventors and attract more attention to their surrounding body of work. Intuitively, less established inventors may have more to gain from the additional exposure associated with a US grant. Therefore, I focus on patent families associated with below-median inventor prominence, as measured by the number of weighted non-focal patents at the time of US filing.[9] Unlike focal patent outcomes that are closely connected to the cohort-specific trends, non-focal outcomes aggregate citations to the inventors' other patents, which span a wide range of filing years and ages. As a result, trends in non-focal citations are shaped more by the maturity of the inventor's broader patent portfolio. So I also include age cohort $\times$ year fixed effects, where the age cohort is based on the average age of the non-focal patents at the time of focal US filing, to account for broader portfolio citation dynamics.

The results show a significant 15.5% increase in citations to non-focal patents following the grant. This effect is primarily driven by increased attention from the US and from third countries, with no meaningful change in citations from the inventors' home country. Third-country citations rise by over 23%, suggesting that the increased exposure associated with a US grant may extend beyond the focal invention to the inventors' broader portfolio. Appendix Figure A7 further decomposes the impact on non-focal citations by quartiles of how established the focal inventors are. The largest gains appear in the bottom quartile (Q1), followed by smaller but still significant effects in Q2; and the estimates are insignificant for Q3 and Q4.[10] These results are consistent with less established inventors benefiting more from the additional exposure associated with US patent grants.

Taken together, the results suggest that US patent grants may broaden the diffusion of

[9]Specifically, for each focal patent, I get a list of inventors and count the number of associated non-focal patents before the US filing of the focal patent that had ever been cited to approximate how established the inventors are. The median value is about 15, equivalent to a focal patent with 3 inventors, where each inventor has, on average, 5 patents receiving citations during the years before the US filing of the focal patent.

[10]Inventor prominence is highly skewed: the $25^{th}$, $50^{th}$, and $75^{th}$ percentiles correspond to approximately 4, 15, and 92 average inventor-weighted non-focal patents, respectively. The $90^{th}$ percentile exceeds 1,300, likely reflecting large firms with very different underlying dynamics; thus, Q4 is capped at the 90th percentile.

foreign-origin knowledge. They facilitate access to third countries, broaden the geographic reach of citations, and draw attention to the inventors' broader portfolio, especially for less established inventors. These patterns are consistent with the US patent system serving as a "platform" for global exposure.

**Robustness Checks**

Appendix Table A2 reports OLS estimates using the same specification as the baseline IV model. The results are very similar in magnitude and direction, suggesting that any selection bias is likely driven by time-invariant characteristics that are already absorbed by the fixed effects. Appendix Tables A3 and A4 show that the estimated effects remain stable across alternative fixed effects, outcome transformations, and instrument constructions, reinforcing the robustness of the findings. Further, Righi and Simcoe (2019) raise concerns that patent applications are assigned nonrandomly to examiners, which may undermine the validity of the examiner-leniency instrument. Appendix Table A5 excludes patents from four of the eight USPTO technology centers: Chemical and Materials Engineering; Mechanical Engineering; Semiconductors, Electrical and Optical; and Biotechnology, where within-art-unit examiner specialization is more pronounced (Righi and Simcoe, 2019). The resulting estimates are similar in magnitude to the baseline results.

Additionally, the first-stage F-statistic is large, consistent with recent studies using examiner-based instruments (Aneja et al., 2024). At the same time, to address concerns that the constructed IV may mechanically overstate the first-stage F-statistic (Hull, 2017), I report Anderson–Rubin Wald statistics to provide weak-instrument-robust inference for Table 2. The significant test results provide additional evidence that weak instruments are unlikely to account for the estimates. Lastly, Appendix Table A6 reports the IV results using an expanded sample of 5,216 patent families whose home-country filing was granted before the US filing, while lifting the restriction that the families have no third-country filings. The results are consistent with the baseline estimates and are slightly larger in percentage terms.

To address potential concerns about heterogeneous treatment timing and dynamic

effects that can bias conventional two-way fixed effects models, I also estimate dynamic group-time average treatment effects using the *csdid* estimator developed by Callaway and Sant'Anna (2021). These results, shown in Appendix Figure A5, display similar post-grant increases and flat pre-trends, providing additional support for the main findings.

## B. Increasing Exposure of Trapped Knowledge

Not all foreign knowledge benefits equally from being granted in the US. Innovations from countries that are already deeply embedded in global innovation networks may need little additional exposure, while those from more peripheral or linguistically isolated countries face greater barriers. Therefore, US grants may play a disproportionately important role in increasing the exposure of knowledge from countries that are more "trapped."

### Language Accessibility

To better understand which frictions constrain the exposure of foreign knowledge, I first examine how the effect of US patent grants on forward citations varies by the linguistic and regional background of the priority country. Table 6 shows that the increase in citations following a US patent grant is primarily driven by Korean and Chinese patents: US grants meaningfully amplify the global exposure of these inventions, both within the US system and across third countries that may otherwise have limited ties to the origin. In contrast, there is little evidence of increased citations for Japanese, English-dominant, or non-English-speaking European patents. This pattern likely reflects their already high levels of exposure due to long-standing integration into global innovation networks. For instance, Japan was a global technological leader during the 1990s, while countries like Germany, France, the UK, and Canada have long-standing institutional, legal, and linguistic alignment with the US and broader international patenting systems. Interestingly, citations from the home country increase only for Chinese patents. This pattern may reflect an additional increase in exposure within the home country itself: where uncertainty about patent quality is greater, the institutional validation associated with a US grant may make the underlying invention more credible and therefore more likely to be considered by domestic Chinese users. These results suggest that US patent grants help elevate innovations from more peripheral or linguistically isolated countries,

highlighting the asymmetric value of US grants in reducing diffusion barriers for otherwise overlooked knowledge.

One possibility is that all the estimated effects of US grants may simply reflect language barriers. Admittedly, language is an important mechanism for knowledge diffusion, and recent evidence on earlier disclosure of English-language technical knowledge from Japan shows that language can shape the transmission of technical information (Higham and Nagaoka, 2026). The evidence below, however, suggests that language alone does not account for the results.

Specifically, I examine citations from countries that share the language of the home country. If the US grant effect were driven only by the availability of an English-language version, we would expect the effect to be muted among citing countries that already share the invention's original language. Table 7 Columns 1 and 2 show that US grants lead to a 17% increase in citations from all countries sharing the home-country language. The effect becomes even larger when excluding the home country itself. Columns 3 and 4 focus on *foreign* countries that share the same language as the home country, and show an increase of approximately 60%. This is informative because many non-English languages in the sample, such as Korean and Japanese, are highly concentrated in a single country; after excluding the home country, the remaining same-language citations are therefore driven largely by cross-country language communities, especially English-speaking countries. The fact that US grants increase citations even within shared-language environments is consistent with US grants increasing exposure along dimensions beyond language accessibility.

### Citation-based Trappedness

To further explore the mechanism, I quantify the extent to which knowledge from a given country and technology class is overlooked by global inventors using a new citation-based measure of "trappedness". This approach is motivated by the distinction between two types of backward citations in patent applications: backward citations added by

applicants and backward citations added by examiners in search reports.[11,12] The key assumption is that examiner-added citations provide a proxy for relevant knowledge that was not readily recognized or brought into the applicant's consideration set at the time of filing. Such knowledge, while formally relevant, may not have contributed directly to the invention itself, while applicant-added citations (mostly in-text) are more likely to signal meaningful technological linkages (Criscuolo and Verspagen, 2008; Alcácer et al., 2009; Cotropia et al., 2013; Verluise et al., 2026). Therefore, a high share of examiner-added citations suggests that relevant prior art from that source is systematically under-recognized by applicants, providing one observable manifestation of "trappedness."

It is important to emphasize that this framework does not assume that examiners are omniscient. Rather, the assumption is that examiners, by virtue of their role to check the originality of the applications and their access to professional search resources, are systematically better positioned than inventors and their patent attorneys in identifying relevant prior art. In this sense, the space of potential citations is less "trapped" with respect to examiners than it is with respect to inventors, making examiner-added citations a useful signal of under-recognized knowledge. This assumption is consistent with prior findings that citations from inventors are much more likely to be from near the inventor's location than examiner-added citations (Alcacer and Gittelman, 2006; Thompson and Fox-Kean, 2005; Thompson, 2006; Singh and Marx, 2013).

As the measure may be confounded by general home bias, which is the tendency of both applicants and examiners to cite patents from their own country due to familiarity or institutional norms (Qiu et al., 2025), I compute the ratio of examiner-added to applicant-added citations. This relative measure helps account for factors that similarly affect both parties, under the assumption that applicants and examiners from the same country share a comparable baseline familiarity with foreign knowledge and face the same institutional environment. Specifically, I calculate this measure using all backward citations made by

[11]There are other types of backward citations, like the citations introduced during an appeal by applicant/proprietor or the citations introduced during opposition. All these other types constitute a minimal share and are excluded from the analysis.

[12]USPTO examiners add backward citations during examination, but do not generate forward citations; all forward citations are created by subsequent patent applications. As a result, forward citation outcomes used in the analysis are not mechanically affected by the granting process and examiner-added backward citations do not contaminate the causal identification of the impact of US patent grants.

international patent filings submitted through the Patent Cooperation Treaty (PCT) to WIPO, which provides a relatively consistent basis for comparison across countries.[13] For each country $c$ being cited, I calculate:

$$Trappedness_c = \log\left(\frac{\sum_i \mathbf{1}\{ExaminerAdded_i = 1\} \times \mathbf{1}\{CitedCountry_i = c\}}{\sum_i \mathbf{1}\{ExaminerAdded_i = 0\} \times \mathbf{1}\{CitedCountry_i = c\}}\right), \quad (4)$$

where $i$ indicates every single citation, and $ExaminerAdded_i$ is an indicator equal to one if this citation is added by the patent examiner. A trappedness measure equal to 2 means that for each citation added by applicants, $e^2 \approx 7.38$ citations are added by examiners.

Recognizing that trappedness may vary not only across countries but also across technology fields, I extend the measure to the country-sector level ($Trappedness_{cj}$), where $j$ denotes the technology class (CPC) of the cited patent. For instance, knowledge in China's engineering and chemistry may be less trapped than knowledge in its social science or health-related fields.

Figure 4 plots the $Trappedness_c$ for all top-cited countries. A higher value indicates greater reliance on examiner intervention relative to applicant awareness. South Korea exhibits the highest trappedness, suggesting that the majority of citations to Korean prior art are added by examiners rather than applicants. Russia, the former Soviet Union, and China also rank high on this measure, while countries such as Germany, Spain, the UK, and Canada are among the least trapped, indicating higher baseline awareness of their knowledge by global inventors.

Table 8 Panel A examines whether the effect of a US grant varies with country-level trappedness. The idea is straightforward: the more a country is trapped, the more likely its innovations are to be overlooked by the broader global audience. If the US grant increases the exposure of such knowledge, then the benefit of US grants should be greater for these more trapped countries. The estimates are consistent with this interpretation. The interaction term between grant status and country-level trappedness is positive and statistically significant, indicating that US patent grants are especially helpful for countries that start from a disadvantaged position. For example, patents

[13]National route filings are excluded because citation tagging practices vary widely across patent offices. For example, the China National Intellectual Property Administration (CNIPA) does not indicate whether a citation was added by the applicant or the examiner.

from the most trapped countries (with $Trappedness_c \simeq 4$) receive 0.182 more normalized citations post-grant, on top of the insignificant baseline effect of 0.0687.

The patterns also differ across citing geographies. Citations from the US show a significantly positive baseline response to grant status, which may reflect the direct increase in exposure within the US patent system and leave less scope for differential effects by country-level trappedness. Further, citations from the home country remain largely unaffected by trappedness, consistent with the intuition that most home inventors are already aware of local knowledge (with China being a notable exception discussed earlier). Notably, the interaction is strongest for citations from third countries, which account for much of the overall relationship. This pattern is consistent with US grants helping inventions reach more distant or disconnected parts of the global innovation network.

One potential limitation is that the results in Panel A may be driven by a few large countries that exhibit high levels of trappedness and account for a sizable share of the sample. To address this, I turn to the more granular country–sector level measure of trappedness: $Trappedness_{cj}$. As shown in Appendix Figure A8, this measure introduces greater variation across country–technology combinations and leads to a more balanced distribution of observations, reducing the risk that the effects are dominated by a few large countries. Panel B of Table 8 presents similar results using this refined measure. These patterns further strengthen the interpretation that US patent grants help elevate under-recognized knowledge: not just from specific countries, but also across a broader set of technologically specialized domains where diffusion barriers are highest.

Together, these findings suggest that US grants are associated with larger increases in diffusion for knowledge that is initially less recognized. The patterns are consistent at both the national and sectoral levels, supporting the interpretation that the additional exposure associated with a US grant may be particularly consequential for more trapped knowledge.

## C. Alternative Mechanisms

The central interpretation is that some patented knowledge remains "trapped" because formal availability does not ensure effective exposure: downstream users may fail

to notice it, recognize its relevance, or confidently assess its value. A US grant may increase effective exposure by making the invention more salient, easier to recognize as relevant, and more credible to downstream innovators, increasing the likelihood that the already-available knowledge is noticed, evaluated, and ultimately built upon. The analyses below examine two channels conceptually distinct from this exposure interpretation: legal and procedural citation incentives, which may increase citations without corresponding technical learning, and commercialization or product-market channels, through which downstream innovators may encounter the invention outside the patent system.

## 1. Legal and Procedural Citation Channels

One alternative interpretation is that the citation response mainly reflects legal or procedural incentives to cite rather than knowledge diffusion: applicants may cite granted US patents to satisfy disclosure obligations or reduce litigation risks, while examiners may cite them to assess novelty, reject overlapping claims, or clarify patent boundaries. Under this interpretation, the grant effect would not necessarily indicate that later inventors build on the underlying technical knowledge; instead, it may reflect strategic acknowledgment of a legally salient US patent.

I first examine this possibility in Table 9. Columns 1 and 2 estimate the effect of US grants on forward citations to the home-priority patent rather than to the US patent or application. These citations are less directly tied to US litigation risk or disclosure strategy because citing the home-country priority document does not identify the operative US patent right in the same way as citing the US patent. If drawing a legal boundary is the goal, then the US patent, rather than the home patent, would probably be cited. The results remain economically large: US grants increase citations to the home-priority patent by approximately 26%. This pattern suggests that the grant increases attention to the underlying invention, not only the legal acknowledgment of the US patent document.

A second legal channel operates through examiners. Once an invention is granted in the US, the patent becomes a more legible and authoritative prior-art reference for later examiners. Examiners may cite it because it helps assess novelty and non-obviousness, reject overlapping claims, or clarify the boundary of patentable subject matter in sub-

sequent applications. Columns 3 and 4 of Table 9 address this examiner-driven channel by excluding examiner-added US citations. This removes the citations most directly associated with examiner search, rejection, and boundary-setting during prosecution. The persistence of the effect among non-examiner-added citations indicates that the estimated grant effect is not driven by examiner behavior.

Finally, if legal and procedural citation channels tied to US patents were the primary mechanism, the citation response should be concentrated among US-origin forward citations, where US prosecution and enforcement concerns are most salient. Instead, Table 3 shows citation increases from the home country and from third countries as well. Taken together, these results suggest that legal and procedural citation channels are unlikely to explain the broader increase in follow-on innovation after US grants.

### 2. Commercialization and Product Market Channels

Another possible channel is that the effect may operate primarily through commercialization rather than the diffusion of previously overlooked technical knowledge. A US patent grant gives the applicant enforceable rights in a large product market, which may facilitate market entry, exports, licensing, or other commercial activity. Downstream innovators may then encounter the invention through products or commercial relationships rather than through the patent document itself. This idea is consistent with evidence that Chinese firms receiving their first successful US patent experience significantly higher export growth (Gong et al., 2023).

To distinguish this channel from increased exposure through the patent system, I restrict the analysis to patent families where the invention is unlikely to generate direct product market interaction. I employ two complementary classification approaches. The first uses the title and abstract of the US patent application: each family in the sample is classified by a large language model (`claude-haiku-4-5`) as either product-oriented innovation describing a physical or digital good that can be directly sold or exported, or non-product-oriented innovation describing a method, process, algorithm, or research tool where no directly tradeable output is named. Appendix Table A7 Panel (a) validates the text-based classification by showing the frequency of keywords that are indicative

of "non-product". The majority of the "non-product" patents are titled with "method". Appendix Table A7 Panel (b) illustrates the classification logic I impose and examples for these "method"-titled patents. The second approach classifies patents by their four-digit CPC technology subclass, independently of patent text. Specifically, I retrieve patent class descriptions from the USPTO[14] and then manually code each subclass as either "product-oriented" or "non-product-oriented" based on these descriptions.

Table 10 reports IV estimates for these non-product subsamples. Across both classification approaches, the effect of a US grant on forward citations remains positive and statistically significant. Using the text-based classification (Columns 1-2), a US grant increases the normalized citation count by 17.4%. The CPC subclass-based estimates (Columns 3-4) are comparable in magnitude, about 20.4%. Since patents in the non-product subsample describe methods, processes, and algorithms rather than directly tradable goods, market entry and export activity are less likely to explain the observed citation growth. Moreover, the dynamics of the effect in Figure 3 show an immediate response following the US grants, while it would take more time if product market interaction is the primary channel. For instance, De Rassenfosse and Zhou (2026) document an average lag from patent priority filing to product release of 6.7 years. Overall, the persistence of the effects, with similar magnitude, in the non-product subsample and the timing of the effect are more consistent with US grants increasing exposure to the underlying knowledge than with a product-market channel.

# V. Conclusion and Discussion

This paper investigates how granting US patents to foreign-origin inventions affects their global diffusion. While the primary function of the US patent system is legal protection, the findings suggest that it may also serve as an influential platform for increasing the effective exposure of already-disclosed foreign knowledge, making it more likely to attract the attention and consideration of downstream innovators.

Notably, US grants are associated with broader diffusion beyond both the United States and the invention's country of origin, increasing citations from third countries

[14]See https://www.uspto.gov/web/patents/classification/cpc/html/cpc.html

and reaching more mid- or low-income countries. They also elevate attention to the inventors' broader portfolio, with the strongest gains observed among less established inventors. Moreover, the diffusion benefits are largest for knowledge originating from countries and sectors that are *ex ante* more trapped. These patterns are consistent with US grants increasing the exposure of knowledge from more disconnected parts of the global innovation system and helping under-recognized ideas enter broader innovation networks.

These findings carry several implications for improving global knowledge diffusion. First, they highlight the unequal landscape of global innovation diffusion, in which knowledge from some countries faces greater barriers to recognition. Second, formal disclosure alone may be insufficient when potential users cannot readily find, understand, or assess foreign knowledge. While the US grant does help overcome some frictions, it cannot fully substitute for more systemic solutions. Reducing informational, legal, and linguistic frictions in the global innovation ecosystem through better citation tools, interoperable metadata, and institutional translation services is critical for ensuring that valuable ideas can travel as freely as their merit deserves.

Finally, this study calls for greater attention to science and technology originating outside the United States. There may be many "hidden diamonds" that either never pursued US patent protection or failed to get a patent due to procedural, linguistic, or institutional barriers, rather than any lack of merit. These ideas may remain under-recognized not because they lack value, but because they never receive sufficient exposure to enter broader innovation networks. Much as Bell et al. (2019) highlight the role of exposure in revealing "lost Einsteins," and Agarwal and Gaule (2020) show the existence of "invisible geniuses", the findings here point to exposure as one force that can untrap "trapped ideas." Expanding efforts to identify, validate, and support high-potential knowledge from underrepresented regions is important not only for equity but also for accelerating global innovation and solving global challenges.

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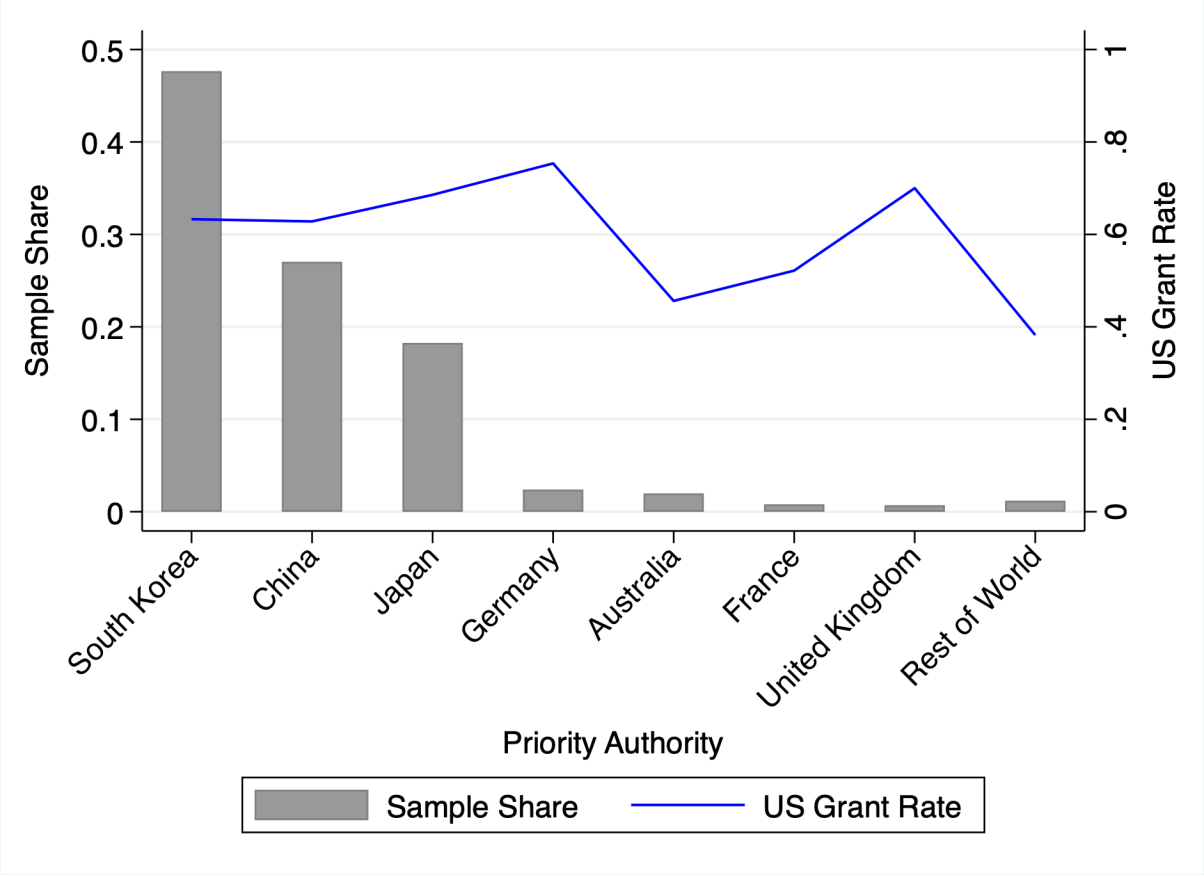
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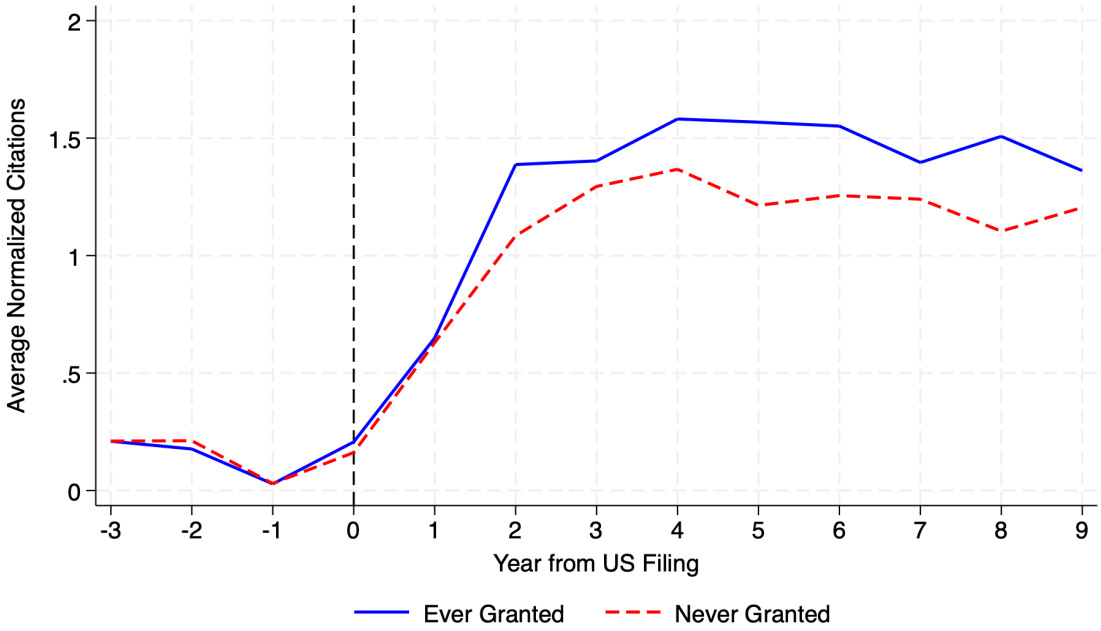
# Main Figures

Figure 1: Distribution of Priority Authorities



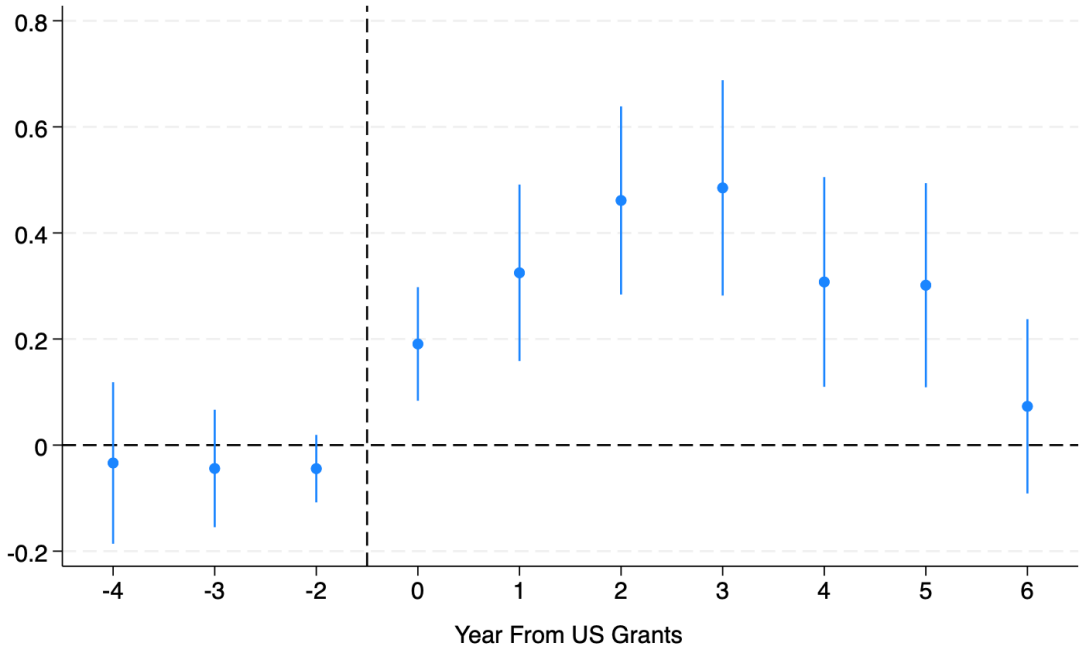

*Note:* This figure shows the share of patent families from different priority authorities (left vertical axis) and the share of patent families whose US filings were granted (right vertical axis) in the sample of interest.

Figure 2: Citation Trends by US Grant Status



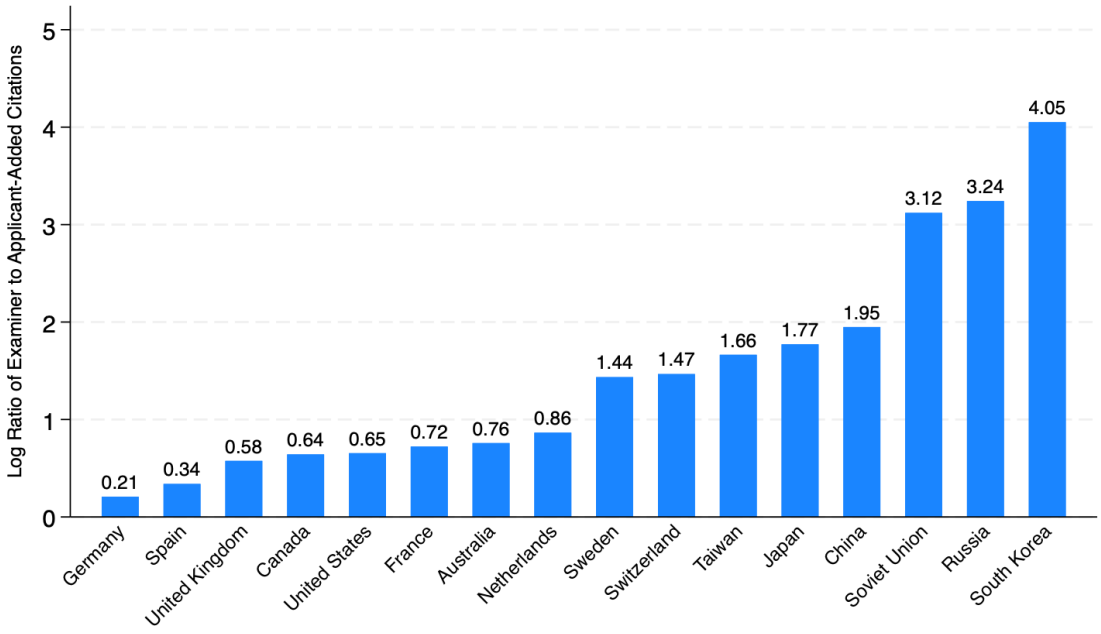

*Note:* This figure shows the average number of normalized citations by year relative to the US filing year, with 3 lead years and 9 lag years, separately for patent families whose US application is granted (blue line) and patent families whose US application is not granted (red line).

Figure 3: Dynamic Effect of US Grants




*Note:* This figure shows an event study plot from an IV regression with 3 lead years and 6 lag years. The year immediately before the US grants is omitted (vertical dashed line). Year $-4$ includes years four or more years before, and year 6 includes years six or more years after. The unit of observation is a patent family-year. Patent family fixed effects and cohort $\times$ year fixed effects are included. Standard errors are clustered by CPC subclass, and the 95% confidence interval is plotted.

Figure 4: Citation-Based Trappedness Measure




*Note:* This figure shows the trappedness ratio of top-cited countries/regions, where trappedness is calculated as the log ratio of the number of backward citations added by examiners to the number of backward citations added by applicants. The higher the log ratio, the more "trapped" the country is. Note that the figure displays this measure for the top-cited countries, although some, such as the Soviet Union, are not present in the sample for analysis.

# Main Tables

Table 1: Summary Statistics

| | *Pre-Grant* ($N = 17,052$) | | | | *Post-Grant* ($N = 13,192$) | | | |
|---|---|---|---|---|---|---|---|---|
| Variable (yearly) | Mean | SD | Min | Max | Mean | SD | Min | Max |
| Any Cite (0/1) | 0.31 | 0.46 | 0 | 1 | 0.53 | 0.50 | 0 | 1 |
| Number of Cites | 0.72 | 1.87 | 0 | 47 | 1.40 | 2.45 | 0 | 70 |
| Any Cite from US (0/1) | 0.21 | 0.41 | 0 | 1 | 0.40 | 0.49 | 0 | 1 |
| Number of Cites from US | 0.45 | 1.57 | 0 | 47 | 0.95 | 2.05 | 0 | 69 |
| Any Cite from Home (0/1) | 0.07 | 0.26 | 0 | 1 | 0.11 | 0.31 | 0 | 1 |
| Number of Cites from Home | 0.11 | 0.54 | 0 | 18 | 0.17 | 0.66 | 0 | 16 |
| Any Cite from Third Countries (0/1) | 0.11 | 0.31 | 0 | 1 | 0.19 | 0.39 | 0 | 1 |
| Number of Cites from Third Countries | 0.16 | 0.55 | 0 | 10 | 0.28 | 0.72 | 0 | 10 |
| Number of Countries | 0.35 | 0.60 | 0 | 4 | 0.64 | 0.75 | 0 | 5 |
| Number of Third Countries | 0.07 | 0.27 | 0 | 3 | 0.14 | 0.38 | 0 | 3 |
| Number of Third High-Income Countries | 0.02 | 0.15 | 0 | 2 | 0.04 | 0.21 | 0 | 3 |
| Number of Third Mid/Low-Income Countries | 0.05 | 0.22 | 0 | 2 | 0.09 | 0.29 | 0 | 2 |
| Average Nonfocal Cites | 1.05 | 1.16 | 0 | 20 | 1.70 | 0.87 | 0 | 12.51 |
| Average Nonfocal Cites from US | 0.53 | 1.01 | 0 | 20 | 0.92 | 0.95 | 0 | 12.29 |
| Average Nonfocal Cites from Home | 0.29 | 0.47 | 0 | 10 | 0.37 | 0.34 | 0 | 3.33 |
| Average Nonfocal Cites from Third Countries | 0.23 | 0.40 | 0 | 15 | 0.41 | 0.31 | 0 | 4 |

*Note:* This table presents summary statistics for the panel, which includes 2,470 patent families, where 888 families are never-treated and 1,582 are treated at some point. "Pre-Grant" includes all never-treated families and the pre-treatment periods of treated families. "Any Cite" is an indicator variable if the patent receives at least one citation in that year. "Average Nonfocal Cites" is the average (weighted) number of citations received by other patents of the focal inventors. For all "Average Nonfocal Cites" measures, summary statistics are presented for patent families with a below-median number of (weighted) nonfocal patents before US filing of the focal patent.

Table 2: US Grant and Follow-on Innovation

| | *Raw Outcomes* | | *Normalized Outcomes* | |
|---|---|---|---|---|
| *DV:* | *Exist Cite (0/1)* | *Num Cite* | *Exist Cite (0/1)* | *Num Cite* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.0715*** | 0.272*** | 0.134*** | 0.203*** |
| | (0.012) | (0.061) | (0.031) | (0.046) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.31 | 0.72 | 0.80 | 0.76 |
| Percent Δ | 23.1% | 37.8% | 16.8% | 26.7% |
| F-Stat | 17795 | 17795 | 17795 | 17795 |
| Weak-IV Robust Wald $\chi^2$ | 37.70 | 20.32 | 19.82 | 20.94 |
| $N$ | 30141 | 30141 | 30141 | 30141 |

*Note:* This table estimates the impact of US patent grants on an indicator of the existence of any forward citation and the number of forward citations. The unit of observation is a patent family-year. Columns 1 and 2 present the raw outcomes, and columns 3 and 4 present the normalized outcomes, where each outcome is divided by the average value of that outcome within the corresponding US filing year and technology class. All estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic and Anderson–Rubin (AR) Wald $\chi^2$-statistic are reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 3: US Grant and Forward Citations by Citing Location

| | *DV: Normalized Number of Citations* | | | | | |
|---|---|---|---|---|---|---|
| *from:* | US | | Home | | Third Countries | |
| | *Exist (0/1)* | *Num* | *Exist (0/1)* | *Num* | *Exist (0/1)* | *Num* |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Post Grant | 0.204*** | 0.226*** | 0.163** | 0.197** | 0.152*** | 0.174*** |
| | (0.043) | (0.059) | (0.078) | (0.087) | (0.053) | (0.065) |
| Patent Family FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Norm. Mean (Pre-Grant) | 0.76 | 0.73 | 0.81 | 0.81 | 0.75 | 0.75 |
| Percent Δ | 26.8% | 31.0% | 20.1% | 24.3% | 20.3% | 23.2% |
| Raw Mean (Pre-Grant) | 0.21 | 0.45 | 0.07 | 0.11 | 0.11 | 0.16 |
| F-Stat | 16357 | 16357 | 16425 | 16425 | 16148 | 16148 |
| *N* | 28218 | 28218 | 23080 | 23080 | 27167 | 27167 |

*Note:* This table estimates the impact of US patent grants by the authority of the citing patents. The unit of observation is a patent family-year. All outcomes are normalized within the corresponding US filing year and technology class, and differences in the number of observations are due to normalization. All estimates are IV estimates. Columns (1) and (2) focus on citations from the US; columns (3) and (4) focus on the home country; and columns (5) and (6) focus on citations from countries other than the US and the home country. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 4: US Grant and Geographic Reach of Innovation

| *DV:* | Num Countries (1) | Num Third Countries (2) | Num High Income Third Countries (3) | Num Mid/Low Income Third Countries (4) |
|---|---|---|---|---|
| Post Grant | 0.123*** | 0.0261*** | 0.00632 | 0.0197*** |
| | (0.017) | (0.007) | (0.004) | (0.006) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.35 | 0.07 | 0.02 | 0.05 |
| Percent Δ | 35.1%*** | 37.3%*** | 31.6% | 39.4%*** |
| F-Stat | 17795 | 17795 | 17795 | 17795 |
| *N* | 30141 | 30141 | 30141 | 30141 |

*Note:* This table estimates the impact of US patent grants on the geographic reach of the patents. The unit of observation is a patent family-year. Countries are classified based on the World Bank Income Classification. All estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 5: US Grant and Increased Inventor Attention

| | *DV: Average Number of Citations to Non-Focal Patents* | | | |
|---|---|---|---|---|
| | *Total* | *From US* | *From Home* | *From Third Countries* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.163*** | 0.0872** | 0.0222 | 0.0539*** |
| | (0.036) | (0.039) | (0.014) | (0.013) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Age Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 1.05 | 0.53 | 0.29 | 0.23 |
| Percent Δ | 15.5%*** | 16.5%** | 7.7% | 23.4%*** |
| F-Stat | 12233 | 12233 | 12233 | 12233 |
| $N$ | 23713 | 23713 | 23713 | 23713 |

*Note:* This table estimates the impact of US patent grants on the attention the home country inventors get. The unit of observation is a patent family-year. The outcomes are the average (weighted) number of citations received by other patents of the focal inventors. The regression includes the set of patent families with a below-median number of (weighted) nonfocal patents before US filing of the focal patent. All estimates are IV estimates. Patent family fixed effects and age cohort × year fixed effects are included, where age cohort is defined by the quintiles of the average age of all non-focal patents at the time of the focal patent's US filing. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 6: US Grant and Follow-on Innovation by Home Country Languages

| *Home Country Language:* | Japanese (1) | Korean (2) | Chinese (3) | Eng.-Dominant (4) | Non-Eng. (EU) (5) |
|---|---|---|---|---|---|
| *DV: Normalized Number of Citations* | | | | | |
| Post Grant | 0.231 | 0.227** | 0.296*** | 0.00680 | 0.327 |
| | (0.210) | (0.095) | (0.102) | (0.250) | (0.211) |
| *DV: Normalized Number of Citations from US* | | | | | |
| Post Grant | 0.317 | 0.225** | 0.204** | 0.0325 | 0.0838 |
| | (0.194) | (0.095) | (0.093) | (0.424) | (0.233) |
| *DV: Normalized Number of Citations from Home* | | | | | |
| Post Grant | 0.204 | 0.132 | 0.437*** | 0.281 | 0.523 |
| | (0.251) | (0.109) | (0.133) | (0.655) | (0.603) |
| *DV: Normalized Number of Citations from Third Countries* | | | | | |
| Post Grant | 0.0341 | 0.236** | 0.174* | -0.325 | -0.0255 |
| | (0.202) | (0.100) | (0.098) | (0.572) | (0.284) |
| Patent Family FE | Yes | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes | Yes |

*Note:* This table estimates the effect of US patent grants on the number of forward citations separately for 5 country groups. The country groups are categorized based on language and regional factors. The "Japanese" group includes Japan; the "Korean" group includes South Korea; and the "Chinese" group encompasses China, including Taiwan province and Hong Kong SAR, where Chinese is the main language. These three languages are separated out because they have a large representation in the sample. The "English-dominant" group consists of countries where English is the predominant language, including the United Kingdom, Canada, Australia, Ireland, Singapore, and New Zealand. The "Non-Eng. (EU)" group includes countries in the European Union, such as Germany, France, Italy, Spain, Netherlands, Sweden, Finland, Denmark, etc. where languages other than English are primarily spoken. The unit of observation is a patent family-year. All outcomes are normalized within the corresponding US filing year and technology class. All estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 7: US Grant and Follow-on Innovation among Same-Language Countries

| | *Normalized Number of Citations from* | | | |
|---|---|---|---|---|
| | *Same-Language Countries* | | *Same-Language-Non-Home Countries* | |
| *DV:* | *Exist (0/1)* | *Num* | *Exist (0/1)* | *Num* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.114* | 0.137* | 0.391* | 0.385* |
| | (0.068) | (0.078) | (0.219) | (0.217) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Norm. Mean (Pre-Grant) | 0.81 | 0.80 | 0.65 | 0.64 |
| Percent Δ | 14.1% | 17.1% | 60.2% | 60.2% |
| Raw Mean (Pre-Grant) | 0.09 | 0.15 | 0.02 | 0.04 |
| F-Stat | 15814 | 15814 | 6760 | 6760 |
| *N* | 25174 | 25174 | 10300 | 10300 |

*Note:* This table estimates the impact of US patent grants on the existence and the number of forward citations from countries sharing the same language as the home country. The unit of observation is a patent family-year. Columns 1 and 2 present citations from countries that share the same language as the home country. Columns 3 and 4 restrict this measure by excluding citations from the home country itself. All outcomes are normalized within the corresponding US filing year and technology class, and all estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 8: US Grant and Follow-on Innovation by Home Trappedness

(a) Country-Level Trappedness

| | *DV: Normalized Number of Citations* | | | |
|---|---|---|---|---|
| | *Total* (1) | *From US* (2) | *From Home* (3) | *From Third Countries* (4) |
| Post Grant | 0.0687 | 0.146** | 0.0497 | -0.0783 |
| | (0.041) | (0.068) | (0.171) | (0.076) |
| Post Grant×Trappedness$_c$ | 0.0454*** | 0.0282 | 0.0426 | 0.0812*** |
| | (0.011) | (0.018) | (0.058) | (0.018) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| *N* | 30212 | 28289 | 23145 | 27238 |

(b) Country-Sector-Level Trappedness

| | *DV: Normalized Number of Citations* | | | |
|---|---|---|---|---|
| | *Total* (1) | *From US* (2) | *From Home* (3) | *From Third Countries* (4) |
| Post Grant | 0.0776 | 0.139* | 0.0767 | 0.0502 |
| | (0.054) | (0.065) | (0.216) | (0.095) |
| Post Grant×Trappedness$_{c,j}$ | 0.0391** | 0.0311* | 0.0309 | 0.0391* |
| | (0.016) | (0.016) | (0.069) | (0.020) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| *N* | 25805 | 24244 | 21141 | 23730 |

*Note:* This table estimates the differential effects of US patent grants by the measure of trappedness at the country level and the country-sector level. The unit of observation is a patent family-year. All outcomes are normalized within the corresponding US filing year and technology class, and differences in the number of observations are due to normalization. All estimates are OLS estimates. Patent family fixed effects and cohort × year fixed effects are included. Standard errors are clustered by priority authority, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 9: US Grant and Follow-on Innovation beyond Legal and Procedural Citation Channels

| | *Normalized Number of Citations* | | | |
|---|---|---|---|---|
| | *to Home Priority Patent* | | *from US Added by Applicants* | |
| *DV:* | *Exist (0/1)* | *Num* | *Exist (0/1)* | *Num* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.213*** | 0.215** | 0.294*** | 0.196** |
| | (0.072) | (0.083) | (0.061) | (0.088) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Norm. Mean (Pre-Grant) | 0.83 | 0.83 | 0.65 | 0.67 |
| Percent Δ | 25.7% | 25.9% | 45.2% | 29.3% |
| Raw Mean (Pre-Grant) | 0.11 | 0.18 | 0.08 | 0.18 |
| F-Stat | 16790 | 16790 | 12456 | 12456 |
| *N* | 25429 | 25429 | 24109 | 24109 |

*Note:* This table estimates the impact of US patent grants on the existence and the number of forward citations under outcomes less directly tied to US legal and procedural citation incentives. Columns 1 and 2 measure citations to the home-country priority patent. Columns 3 and 4 measure citations from US patents that were added by applicants. All outcomes are normalized within the corresponding US filing year and technology class, and all estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table 10: US Grant and Follow-on Innovation with Non-Product Patent Families

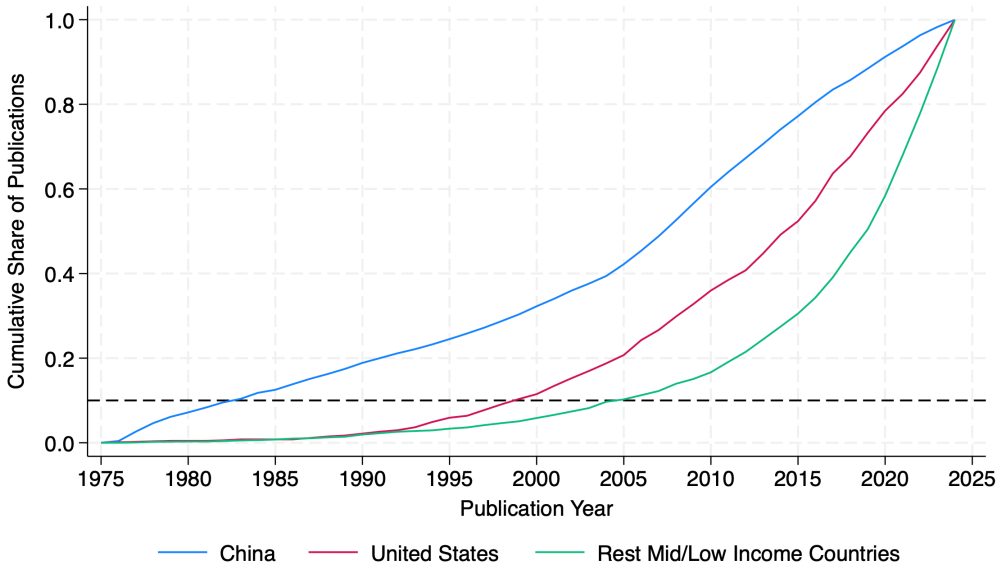
| *Classification based on:* | Title & Abstract | | CPC Subclass | |
|---|---|---|---|---|
| | *Normalized Num Citations* | | *Normalized Num Citations* | |
| *DV:* | *Exist (0/1)* | *Num* | *Exist (0/1)* | *Num* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.178*** | 0.134* | 0.111** | 0.155* |
| | (0.054) | (0.070) | (0.054) | (0.086) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Norm. Mean (Pre-Grant) | 0.80 | 0.77 | 0.80 | 0.76 |
| Percent Δ | 22.3% | 17.4% | 13.9% | 20.4% |
| Raw Mean (Pre-Grant) | 0.32 | 0.81 | 0.33 | 0.80 |
| F-Stat | 4566 | 4566 | 10275 | 10275 |
| *N* | 7086 | 7086 | 9614 | 9614 |

*Note:* This table estimates the impact of US patent grants on the existence and the number of forward citations, focusing on a subset of patent families where the technology is not directly related to a product. The unit of observation is a patent family-year. Columns 1 and 2 use non-product patent families classified based on titles and abstracts. Columns 3 and 4 use non-product patent families classified based on CPC subclass. All outcomes are normalized within the corresponding US filing year and technology class, and all estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

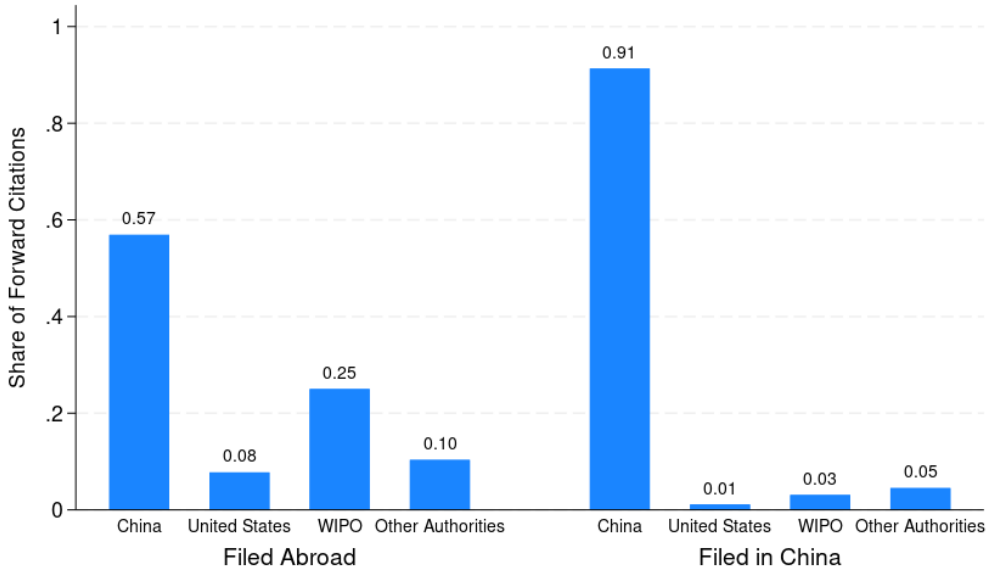
# Appendix Figures and Tables

Figure A1: Hybrid Rice Publications and Patents

(a) Cumulative Distribution of Publications by Region

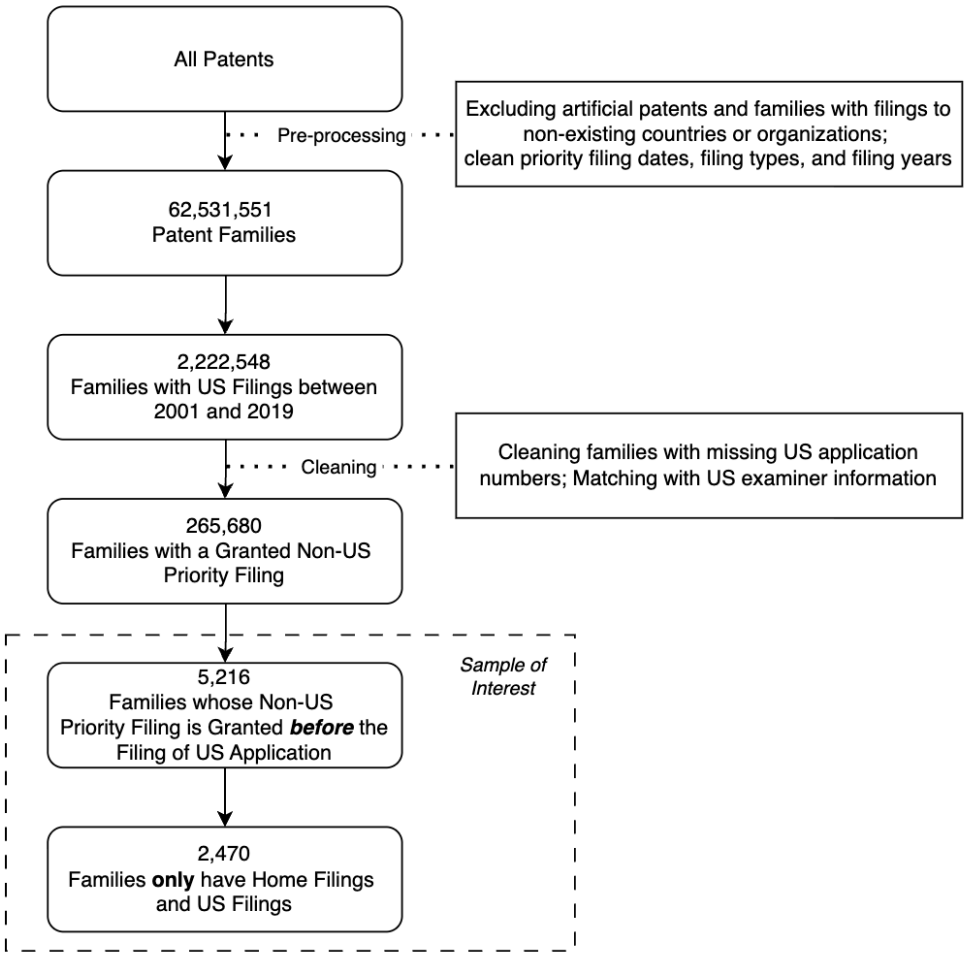



(b) Forward Citation Distribution of Chinese Rice Patents



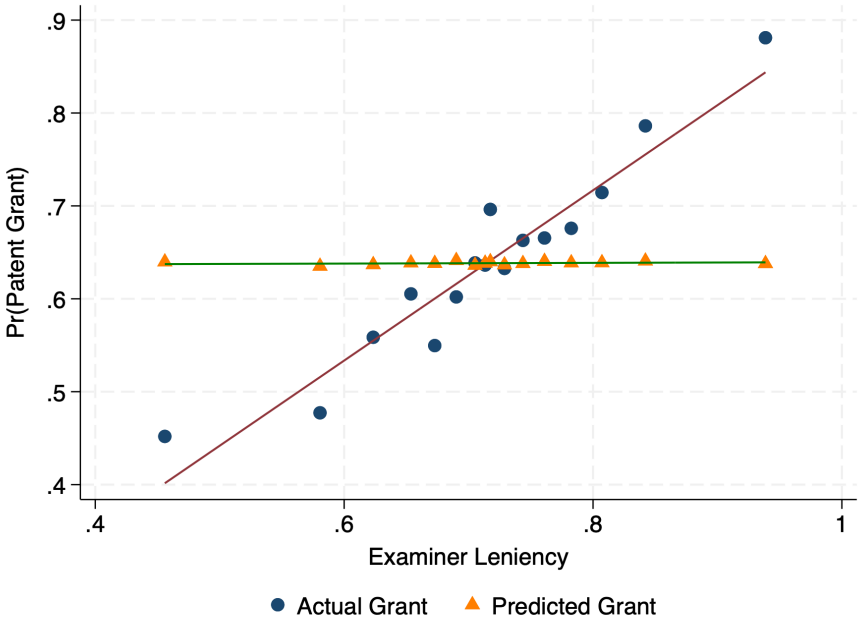

*Note:* This figure focuses on hybrid rice publications and patents. Panel (a) shows the cumulative distribution of the number of hybrid rice publications over time for China, the US, and other mid- or low-income countries. The publications are retrieved from China National Knowledge Infrastructure (CNKI) for China and from Dimensions.org for US and other mid- or low-income countries. Panel (b) shows the distribution of forward citations received by all hybrid rice patents authored by Chinese researchers, separately by whether the patent is filed in China or abroad, and by the jurisdictions where the forward citations are made (China, US, WIPO, and others). China is a global leader in hybrid rice: out of a total of 3,737 hybrid rice patents, 85.4% are from China. Among these, 122 patents are filed in a foreign country, primarily in the US. Conditional on ever being cited, the patents filed domestically receive, on average, 4.8 forward citations, while patents filed in the US receive 8.3 forward citations on average.

Figure A2: Data Construction Flow Chart



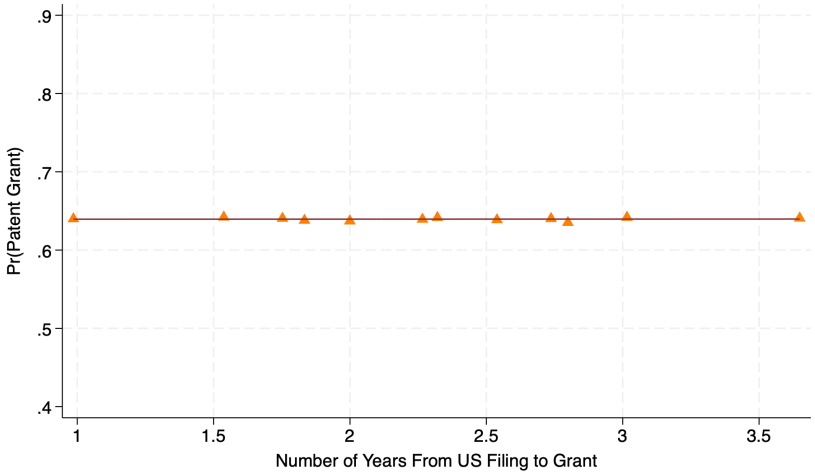

*Note:* This figure shows the sample construction process beginning with all the patent families retrieved from PATSTAT Global via MIT Libraries Dataverse.

Figure A3: Probability of Patent Grant by Examiner Leniency



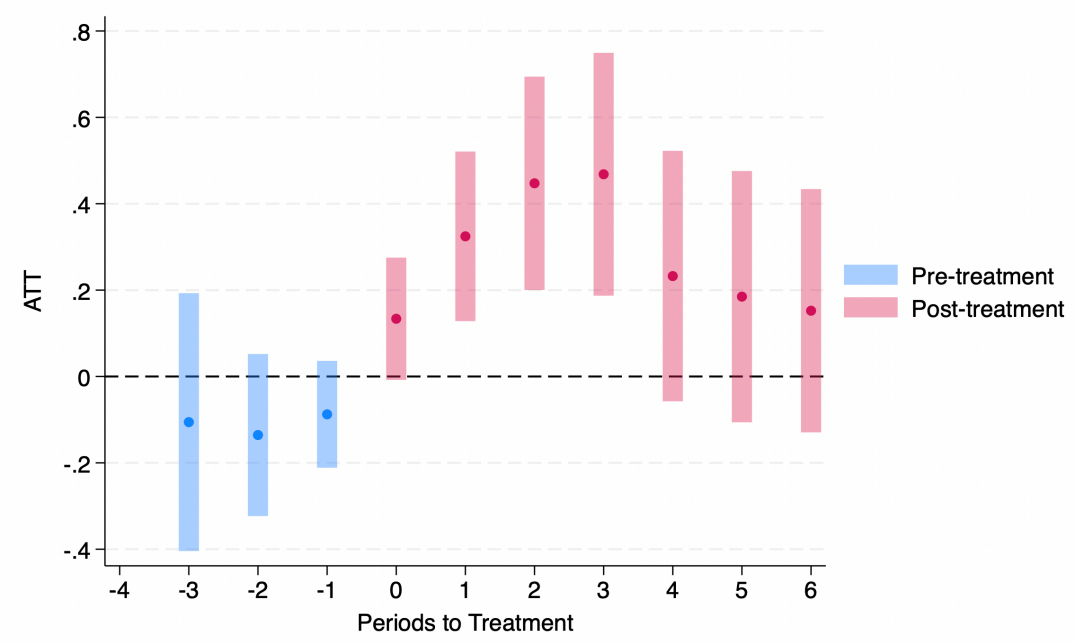

*Note:* This figure relates the examiner leniency measure, residualized by priority authority, to two variables: (1) the patent grant rate, shown in blue, and (2) the predicted patent grant rate, shown in orange. The predicted patent grant probability is based on observable characteristics that may be correlated with patent value and are fixed at the time of application, including the sector, year, the priority authority, the number of inventors, and the size of the patent family.

Figure A4: Probability of Patent Grant and Timing of Patent Grant



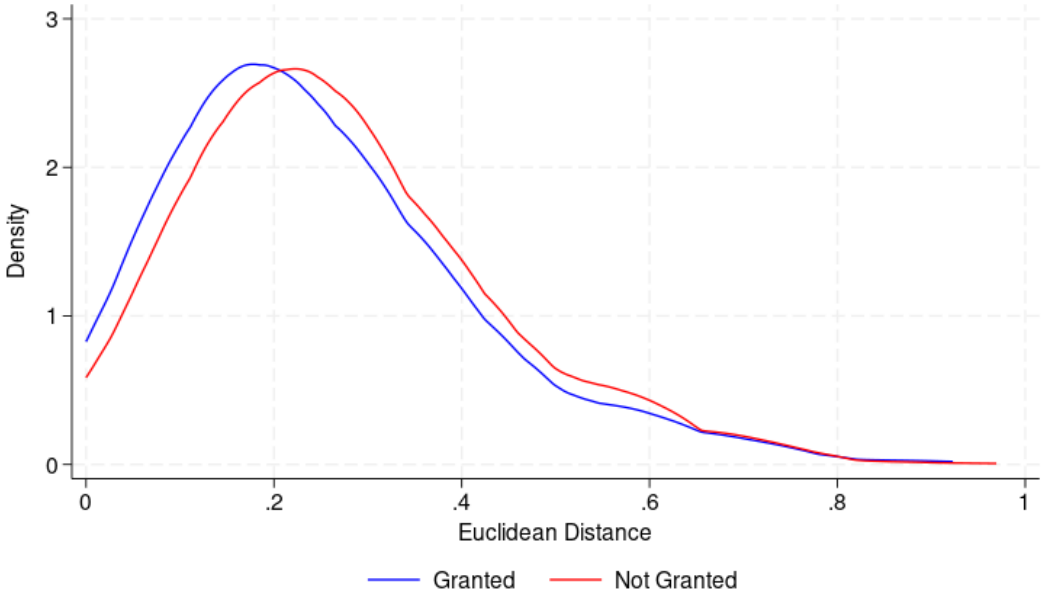

*Note:* This figure relates the lag between US patent grant dates and patent filing dates, residualized by priority authority, to the predicted patent grant rate. The predicted patent grant probability is based on observable characteristics that may be correlated with patent value and are fixed at the time of application, including the sector, year, the priority authority, the number of inventors, and the size of the patent family.

Figure A5: Additional Panel Event Study (Callaway and Sant'Anna, 2021)




*Note:* This figure shows the event study plot following Callaway and Sant'Anna (2021) with 3 lead years and 6 lag years. The years before the US patent grant are marked blue, and the years after the US grant are marked red. The unit of observation is a patent family-year. 95% confidence interval is plotted.

Figure A6: Distance between Citing and Cited Sectors by Grant Status

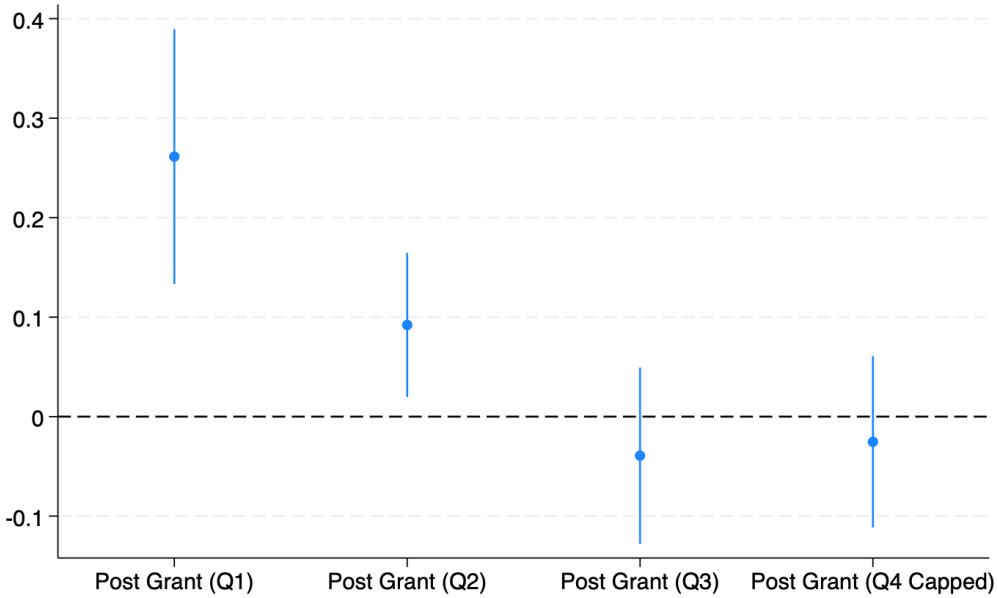



*Note:* This figure plots the distribution of the Euclidean distance between the citing and cited country-classes' technological profiles, separately for patent families whose US application is granted (in blue) and those that are not granted (in red). For each citation, the distance is calculated at the country-pair-class-pair level. For citing country A with CPC class X and cited country B with CPC class Y, I first represent each country–class combination as an N-dimensional vector of CPC subclass shares (where N is the total number of 4-digit CPC subclasses under the 3-digit CPC classes X and Y), where the elements corresponding to subclasses under that class sum to one and all other elements are zero. To compare both sectors between the two countries, I combine information from both classes within each country by summing the two relevant vectors, yielding one aggregated vector for the citing side $(A, X+Y)$ and one for the cited side $(B, X+Y)$. The Euclidean distance between these two vectors is then computed as:

$$d = \sqrt{\sum_{i=1}^{N} \left(v_i^A - v_i^B\right)^2},$$

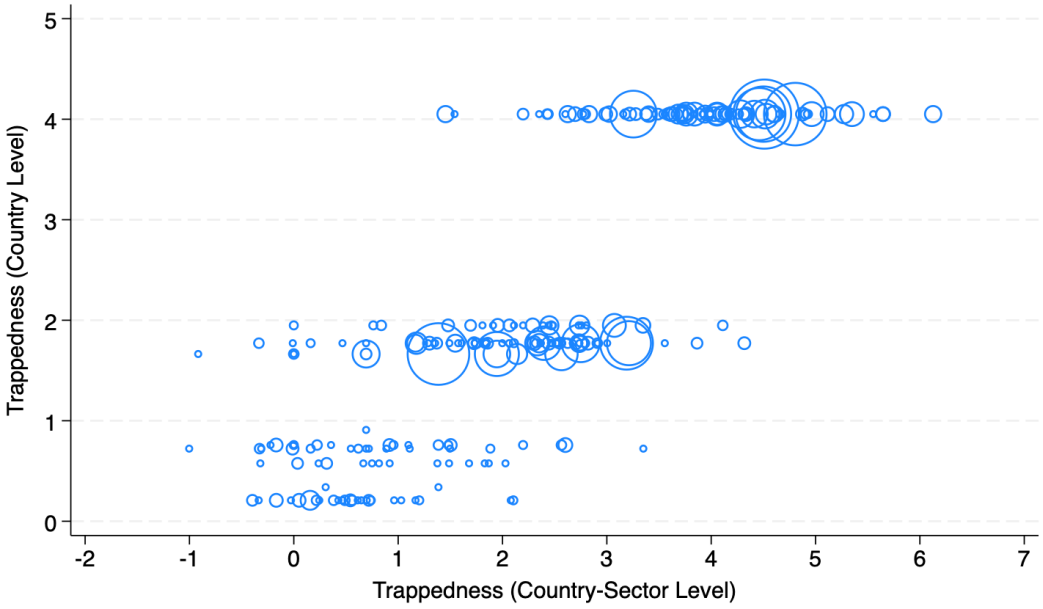
where $v_i^A$ and $v_i^B$ denote the shares for subclass $i$ in the aggregated citing and cited vectors, respectively. A smaller distance indicates that the citing and cited countries are concentrated in similar subclasses, suggesting technological proximity, while a larger distance reflects specialization in different subclasses.

Figure A7: US Grant and Attention Diffusion – Subsample by Pre-filing Stock




*Note:* This figure shows the impact of US patent grants on the attention the home country inventors get, separately for 4 groups of focal patents varying in how established the inventors are. The unit of observation is a patent family-year. The outcome is the average (weighted) number of citations received by other patents of the focal inventors. Patent family fixed effects and age cohort × year fixed effects are included, where age cohort is defined by the quintiles of the average age of all non-focal patents at the time of the focal patent's US filing. Inventor prominence is highly skewed: the $25^{th}$, $50^{th}$, and $75^{th}$ percentiles correspond to approximately 4, 15, and 92 average inventor-weighted non-focal patents, respectively. The 90th percentile exceeds 1,300, likely reflecting large firms with very different underlying dynamics; thus, Q4 is capped at the 90th percentile. Standard errors are clustered by CPC subclass, and the 95% confidence interval is plotted.

Figure A8: Trappedness Measures at Country and Country-Sector Level




*Note:* This figure shows a scatter plot between $Trappedness_c$ and $Trappedness_{c,j}$. The size of the circles is proportional to the number of patent families in the specific country-sector in the sample.

Table A1: Distribution of CPC Class (3-digit)

| CPC Class | Frequency | Percent | Short Description |
|---|---|---|---|
| H01 | 440 | 15.31% | Basic electric elements |
| G06 | 408 | 14.20% | Computing; Calculators |
| H04 | 258 | 8.98% | Electric communication |
| G01 | 213 | 7.41% | Measuring; Testing |
| A61 | 153 | 5.32% | Medical or veterinary science |
| A63 | 126 | 4.38% | Sports; Games |
| G02 | 70 | 2.44% | Optical devices |
| H03 | 57 | 1.98% | Basic electronic circuits |
| B60 | 55 | 1.91% | Vehicles in general |
| H02 | 54 | 1.88% | Generation of electricity |
| G03 | 42 | 1.46% | Photographic technology |
| G11 | 42 | 1.46% | Information storage |
| B01 | 39 | 1.36% | Physical or chemical processes |
| B41 | 38 | 1.32% | Printing |
| G09 | 38 | 1.32% | Education; Display devices |
| F16 | 37 | 1.29% | Engineering elements |
| A47 | 33 | 1.15% | Furniture; Domestic articles |
| H05 | 31 | 1.08% | Other electric techniques |

*Note:* This table shows the frequency of top CPC classes in the panel sample.

Table A2: US Grant and Follow-on Innovation – Ordinary Least Squares

| | *Raw Outcomes* | | *Normalized Outcomes* | |
|---|---|---|---|---|
| *DV:* | *Exist Cite (0/1)* | *Num Cite* | *Exist Cite (0/1)* | *Num Cite* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.0682*** | 0.260*** | 0.122*** | 0.196*** |
| | (0.011) | (0.056) | (0.031) | (0.048) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.31 | 0.72 | 0.80 | 0.76 |
| Percent Δ | 22.0% | 36.1% | 15.3% | 25.8% |
| *N* | 30235 | 30235 | 30235 | 30235 |

*Note:* This table estimates the impact of US patent grants on an indicator of the existence of any forward citation and the number of forward citations. The unit of observation is a patent family-year. Columns 1 and 2 present the raw outcomes, and columns 3 and 4 present the normalized outcomes, where each outcome is divided by the average value of that outcome within the corresponding US filing year and technology class. All estimates are OLS estimates. Patent family fixed effects and cohort × year fixed effects are included. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A3: US Grant and Follow-on Innovation – Baseline Robustness Checks

| | Alternative FE | | Alternative DV | | Alternative *Leniency* | |
|---|---|---|---|---|---|---|
| | | | *Log(y+ϵ)* | *Asinh(y)* | *Unadjusted* | *Examiner-Year* |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Post Grant | 0.333*** | 1.023*** | 1.083*** | 0.146*** | 0.200*** | 0.205*** |
| | (0.071) | (0.037) | (0.174) | (0.022) | (0.045) | (0.046) |
| Patent Family FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | | Yes | | | | |
| Cohort × Year FE | Yes | | Yes | Yes | Yes | Yes |
| Country × Tech Class × Year FE | Yes | | | | | |
| F-Stat | 11630 | 23648 | 17795 | 17795 | 18849 | 26689 |
| *N* | 21589 | 30150 | 30141 | 30141 | 30235 | 30210 |

*Note:* This table estimates the impact of the US patent grants with alternative specifications. The unit of observation is a patent family-year. Columns 1 and 2 present alternative fixed effects. Columns 3 and 4 present alternative dependent variable transformations. Columns 5 and 6 present alternative instrumental variable adjustments. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A4: US Grant and Follow-on Innovation - Alternative IV

| | *Raw Outcomes* | | *Normalized Outcomes* | |
|---|---|---|---|---|
| *DV:* | *Exist Cite (0/1)* | *Num Cite* | *Exist Cite (0/1)* | *Num Cite* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.137*** | 0.294** | 0.331*** | 0.405*** |
| | (0.026) | (0.147) | (0.078) | (0.133) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.31 | 0.72 | 0.80 | 0.76 |
| Percent Δ | 44.19% | 40.83% | 41.38% | 53.29% |
| F-Stat | 1172 | 1172 | 1172 | 1172 |
| Weak-IV Robust Wald $\chi^2$ | 30.86 | 4.33 | 19.58 | 10.19 |
| $N$ | 30235 | 30235 | 30235 | 30235 |

*Note:* This table estimates the impact of US patent grants on an indicator of the existence of any forward citation and the number of forward citations. The unit of observation is a patent family-year. Columns 1 and 2 present the raw outcomes, and columns 3 and 4 present the normalized outcomes, where each outcome is divided by the average value of that outcome within the corresponding US filing year and technology class. All estimates are IV estimates based on $\widehat{Post}_{it} \times Leniency_i$. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic and Anderson–Rubin (AR) Wald $\chi^2$-statistic are reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A5: US Grant and Follow-on Innovation - Excluding Specialized Tech Centers

| | *Raw Outcomes* | | *Normalized Outcomes* | |
|---|---|---|---|---|
| *DV:* | *Exist Cite (0/1)* | *Num Cite* | *Exist Cite (0/1)* | *Num Cite* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.0889*** | 0.178* | 0.174*** | 0.176** |
| | (0.028) | (0.100) | (0.065) | (0.087) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.33 | 0.79 | 0.81 | 0.77 |
| Percent Δ | 26.9% | 22.5% | 21.5% | 22.9% |
| F-Stat | 20268 | 20268 | 20268 | 20268 |
| *N* | 10412 | 10412 | 10412 | 10412 |

*Note:* This table estimates the impact of US patent grants on an indicator of the existence of any forward citation and the number of forward citations excluding 4 centers where strong within-art unit examiner specialization may undermine the validity of the examiner instrument (Righi and Simcoe, 2019). The unit of observation is a patent family-year. Columns 1 and 2 present the raw outcomes, and columns 3 and 4 present the normalized outcomes, where each outcome is divided by the average value of that outcome within the corresponding US filing year and technology class. All estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A6: US Grant and Follow-on Innovation - Expanded Sample

| | *Raw Outcomes* | | *Normalized Outcomes* | |
|---|---|---|---|---|
| *DV:* | *Exist Cite (0/1)* | *Num Cite* | *Exist Cite (0/1)* | *Num Cite* |
| | (1) | (2) | (3) | (4) |
| Post Grant | 0.0645*** | 0.299*** | 0.135*** | 0.212*** |
| | (0.008) | (0.042) | (0.025) | (0.031) |
| Patent Family FE | Yes | Yes | Yes | Yes |
| Cohort × Year FE | Yes | Yes | Yes | Yes |
| Mean (Pre-Grant) | 0.28 | 0.65 | 0.77 | 0.73 |
| Percent Δ | 23.0% | 46.0% | 17.5% | 29.0% |
| F-Stat | 21801 | 21801 | 20247 | 20247 |
| $N$ | 59261 | 59261 | 57630 | 57630 |

*Note:* This table estimates the impact of US patent grants on an indicator of the existence of any forward citation and the number of forward citations using the expanded sample of 5,216 patent families. The unit of observation is a patent family-year. Columns 1 and 2 present the raw outcomes, and columns 3 and 4 present the normalized outcomes, where each outcome is divided by the average value of that outcome within the corresponding US filing year and technology class. All estimates are IV estimates. Patent family fixed effects and cohort × year fixed effects are included. Kleibergen-Paap weak identification F-statistic is reported. Standard errors are clustered by CPC subclass, and *, **, and *** indicate significance at the 10%, 5%, and 1% levels.

Table A7: Validation of LLM Classification Logic

(a) Frequency of Top Keywords Indicative of Non-Product Patents

| Keyword | Non-product (%) | Product (%) | Ratio |
|---|---|---|---|
| *method* | 94.6 | 36.6 | 2.6× |
| *processing* | 11.7 | 3.8 | 3.1× |
| *medium* | 10.4 | 2.5 | 4.2× |
| *information* | 10.4 | 2.6 | 4.0× |
| *program* | 8.5 | 2.2 | 3.8× |

(b) Classification of "Method" Patents: Product vs. Non-Product

| Title pattern | Classification logic | Example titles |
|---|---|---|
| *Panel A: classified as* ***product*** *(is_product = 1)* | | |
| *"Method and [a product] for X"* | A physical device named alongside the method | *Method and apparatus for cooling control* |
| *"Method of making/fabricating [tradeable good]"* | Output is a directly tradeable physical good | *Method of weaving camouflage fabric; Method of fabricating flat panel display* |
| *Panel B: classified as* ***non-product*** *(is_product = 0)* | | |
| *"Method for detecting/calibrating"* | Output is information or an internal metric | *Method for detecting metal contamination*; *Method for calibrating semiconductor tester* |
| *"Method for providing/controlling [software function]"* | Pure software or algorithm —no physical product | *Method for providing virtual space*; *Method for controlling network traffic* |
| *"Method of forming/processing [intermediate]"* | Manufacturing steps with no named tradeable output | *Method of forming a vertical image sensor*; *Method for processing ceramic powder suspension* |

*Notes:* This table validates the product classification. Panel (a) shows the top keywords in patents classified as non-product and the share of patents in each class whose title contains the keyword. Panel (b) shows the classification logic for "method"-titled patents in the 2,470-family analysis sample. The key distinction is whether the patent names a tangible tradeable output (product = 1) or describes a process, algorithm, or technique with no directly tradeable end-good (product = 0). Classification uses `claude-haiku-4-5` on the title and abstract.